

PRACTICAL NO 3

DATA VISUALIZATION WITH POWERBI AND TABLEAU

Lesson 3: Creating Advanced Visualizations in Tableau

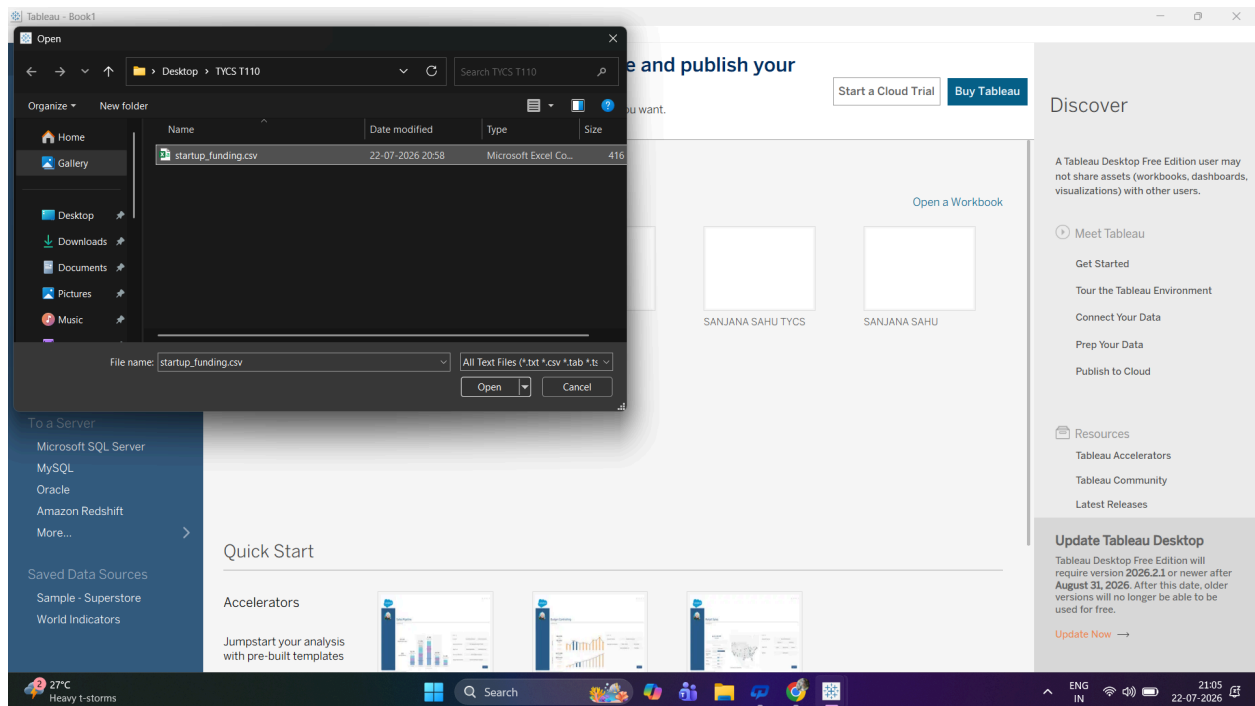
Dr. Mahendra Kanojia

- a) Create stacked and grouped bar charts for comparison
 - b) Build treemaps and pie charts for part-to-whole analysis
 - c) Design scatter plots to identify relationships
 - d) Create dual-axis and combination charts
 - e) Develop histograms for distribution analysis
 - f) Build box-and-whisker plots
 - g) Use Gantt charts and advanced date visualizations
- Dataset: Indian Startup Funding

Step A: Creating Stacked and Grouped Bar Charts for Comparison

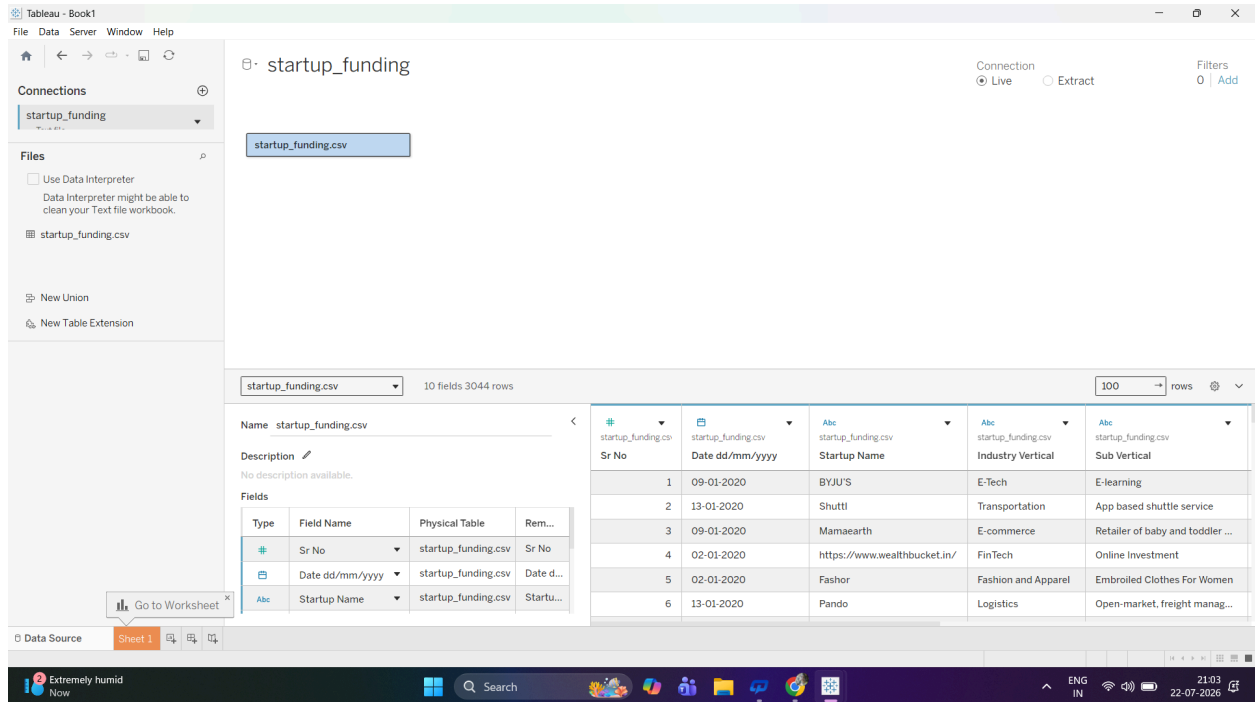
1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your startup_funding.csv, and click Open.

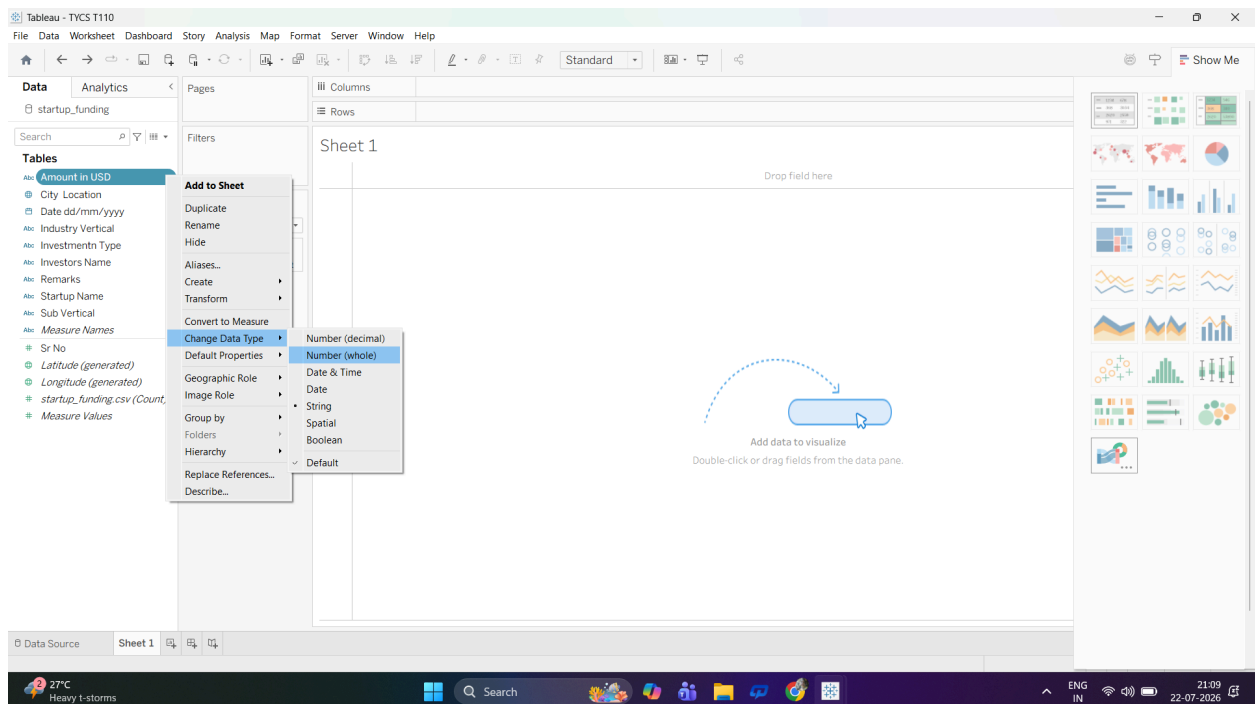


2. Click on Sheet 1 at the bottom-left corner to enter your workspace.

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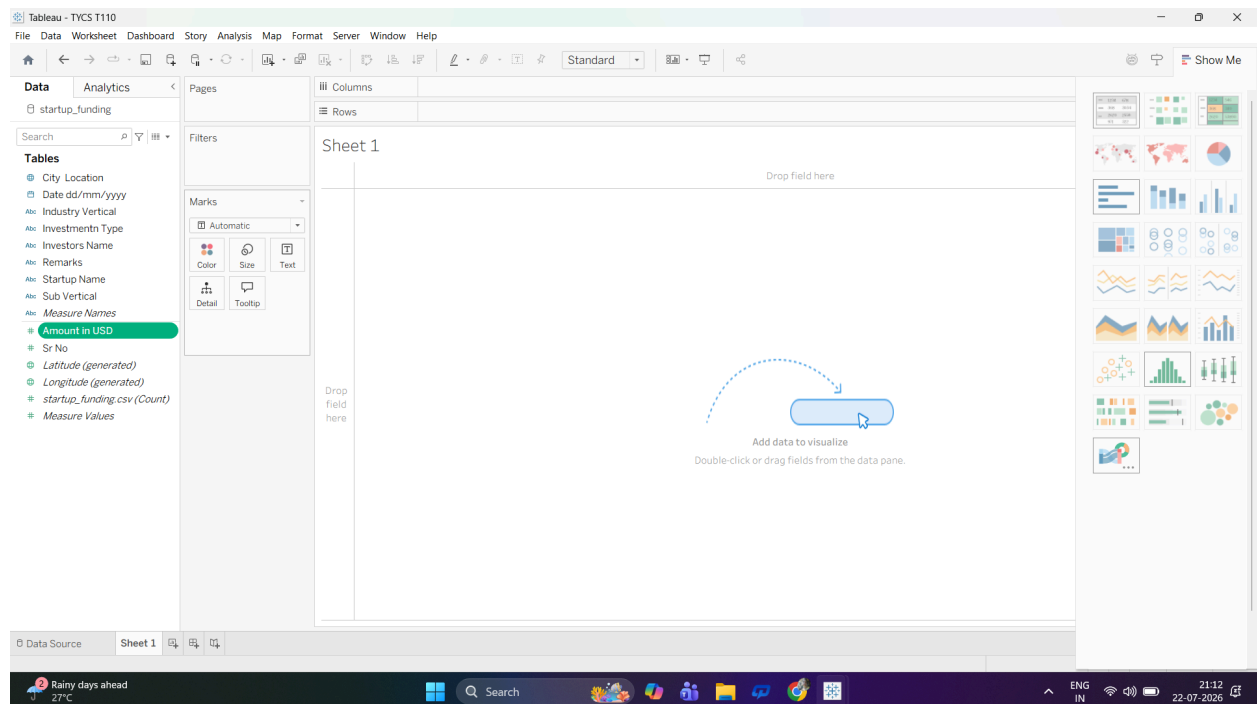
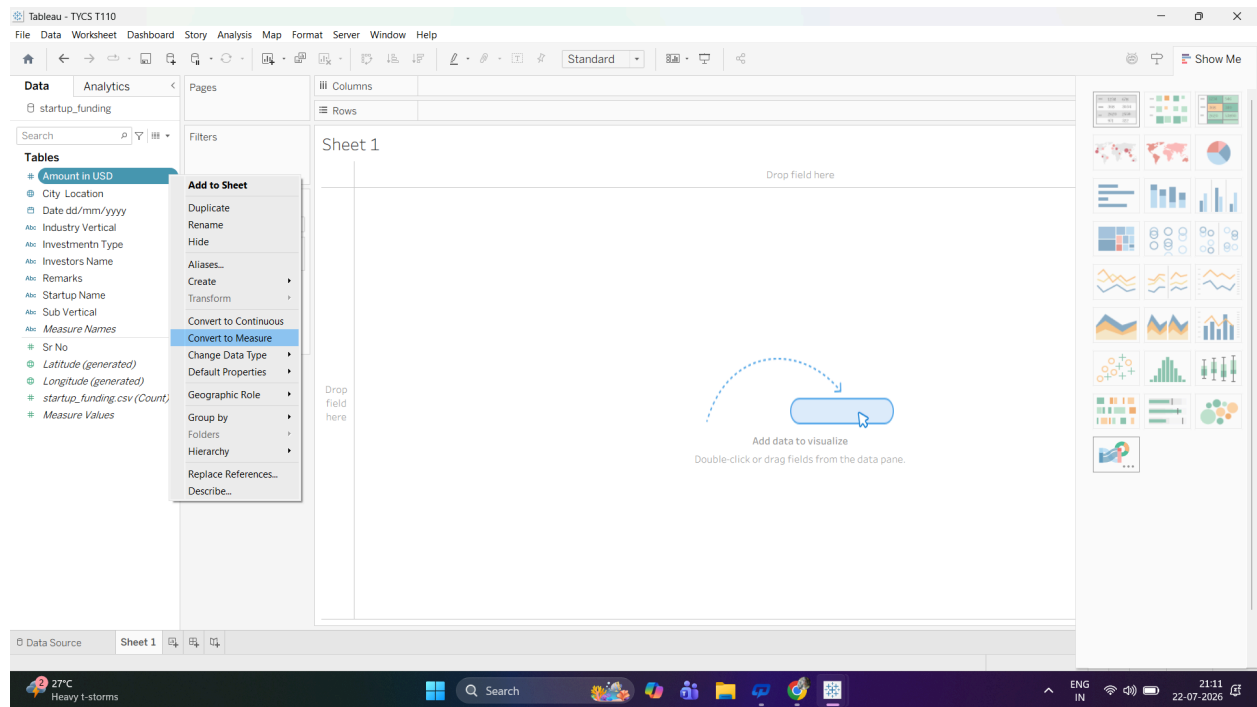


3. Fix the Data Type: In the Data pane on the left, locate Amount in USD. Click the small Abc text icon next to it and select Number (Whole) from the dropdown menu to convert it into numeric data.



4. Convert to Measure: Right-click Amount in USD in the Data pane and select Convert to Measure. It will turn green and move to the bottom half of the pane.

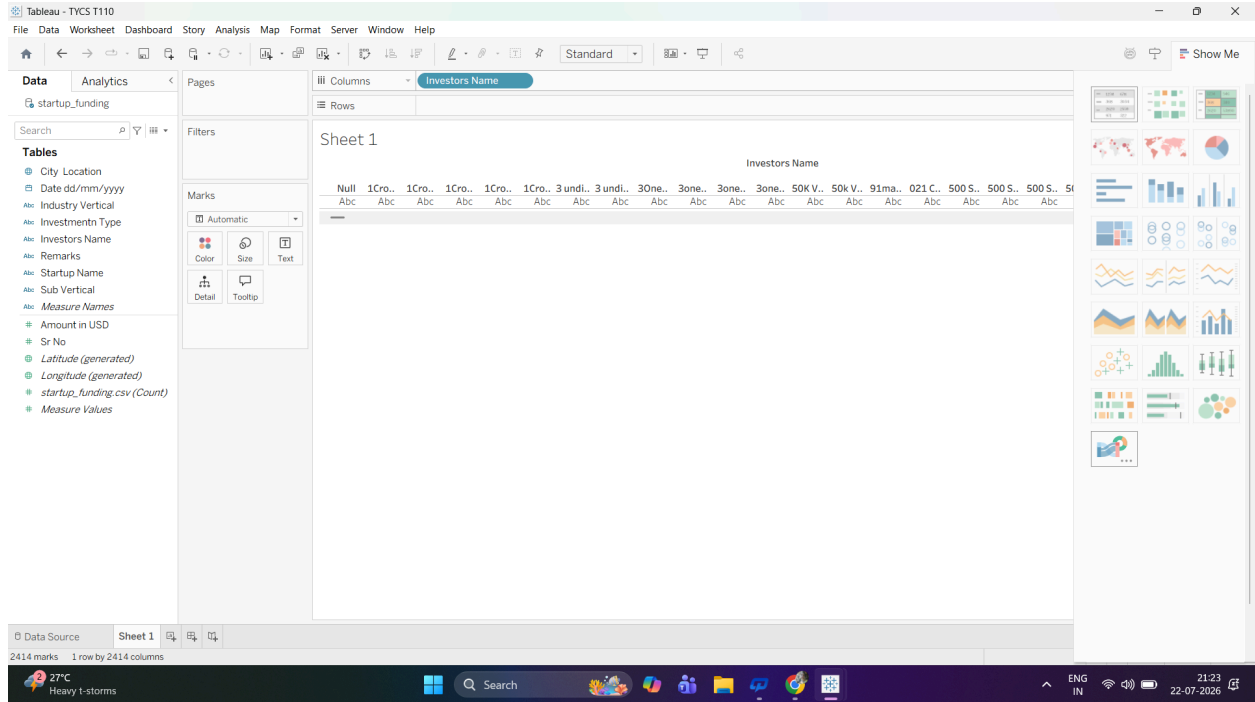
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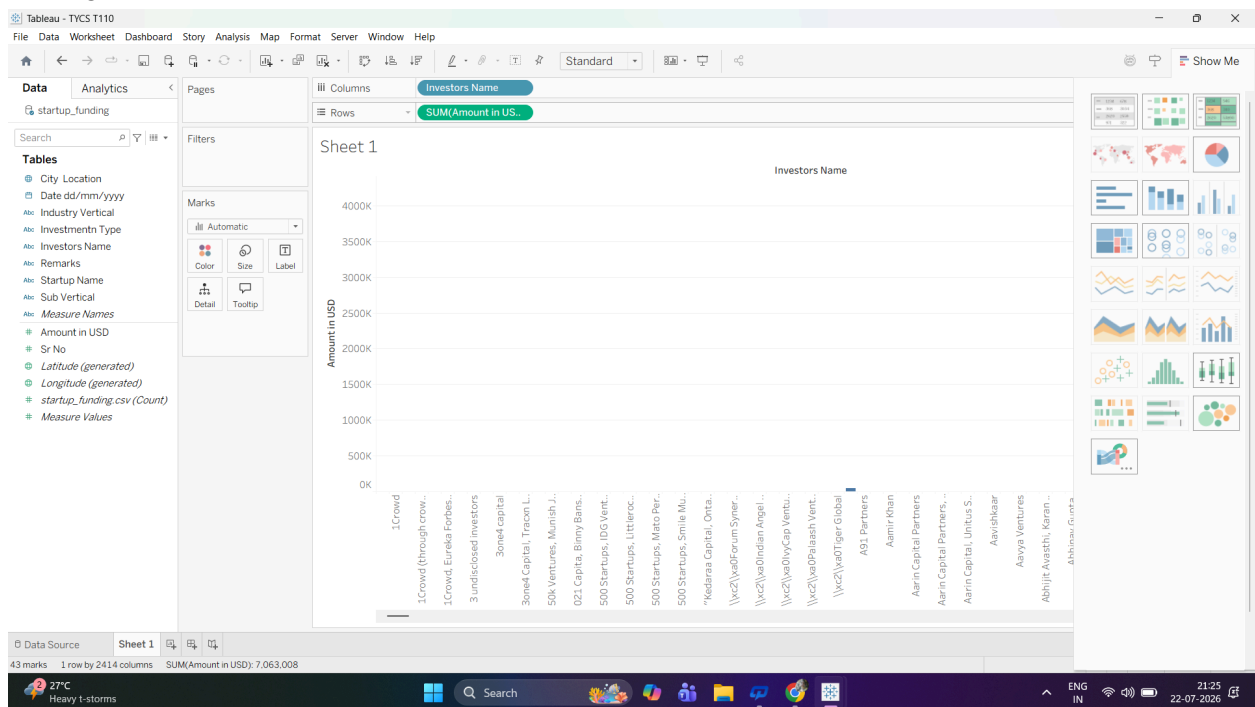
2. Building a Stacked Bar Chart

1. From the Data Pane on the left, click and drag the InvestmentnType dimension onto the Columns shelf.

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2. Drag the Amount in USD measure onto the Rows shelf.



3. Take the City Location dimension from the Data Pane and drop it directly onto the Color tile in the Marks card.

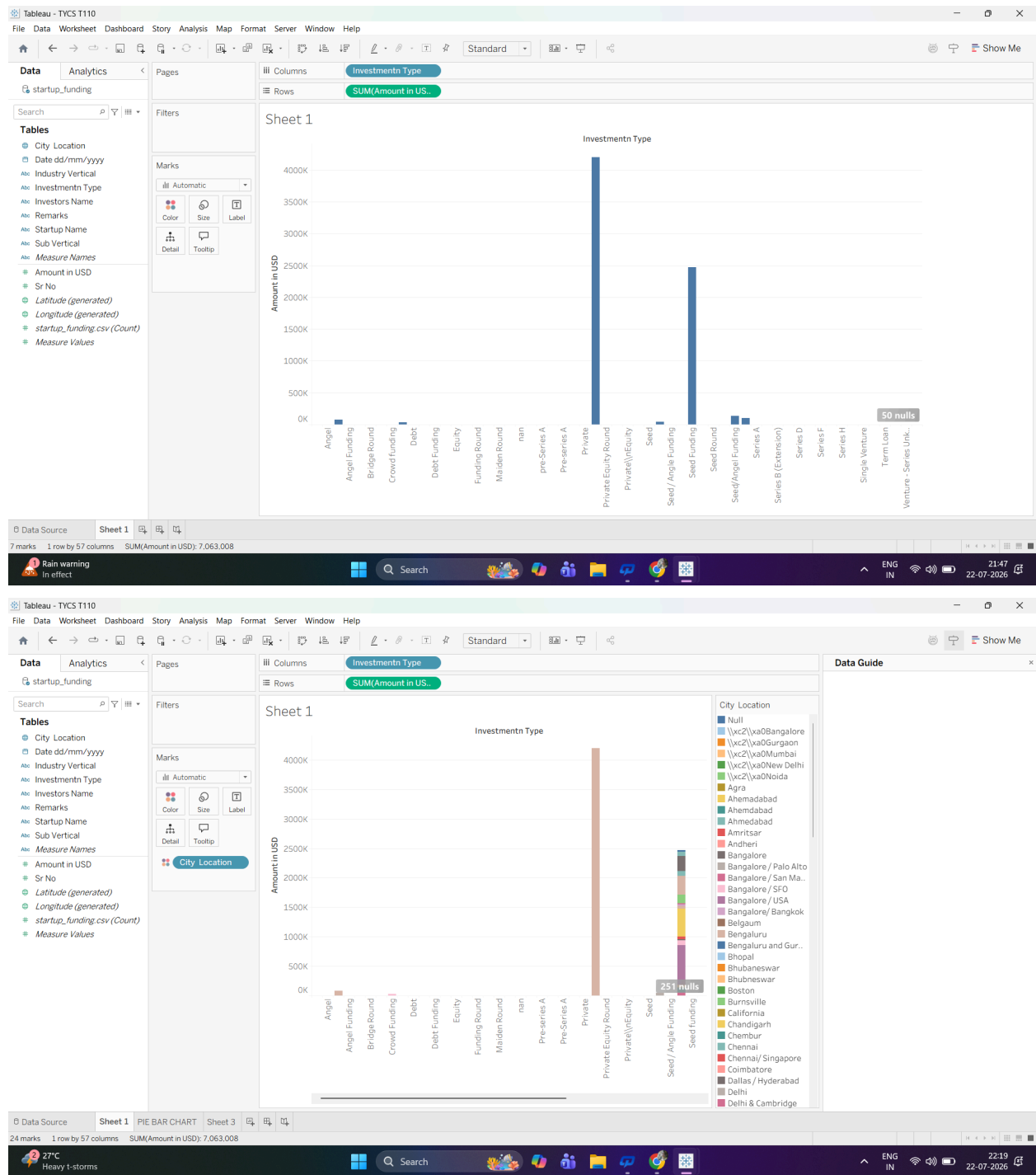
3. Converting to a Grouped Bar Chart (Side-by-Side)

1. Leave Amount in USD on the Rows shelf.

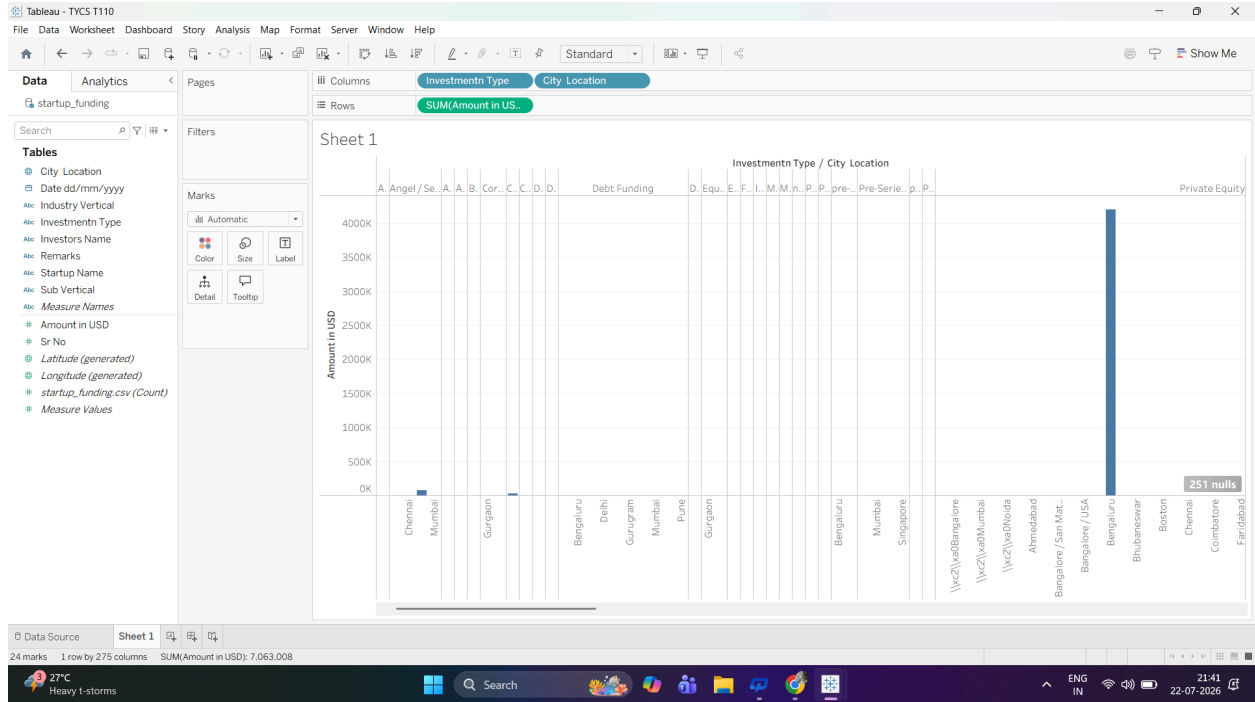
3. Take the City Location dimension from the Data Pane and drop it directly onto the

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Color tile in the Marks card.



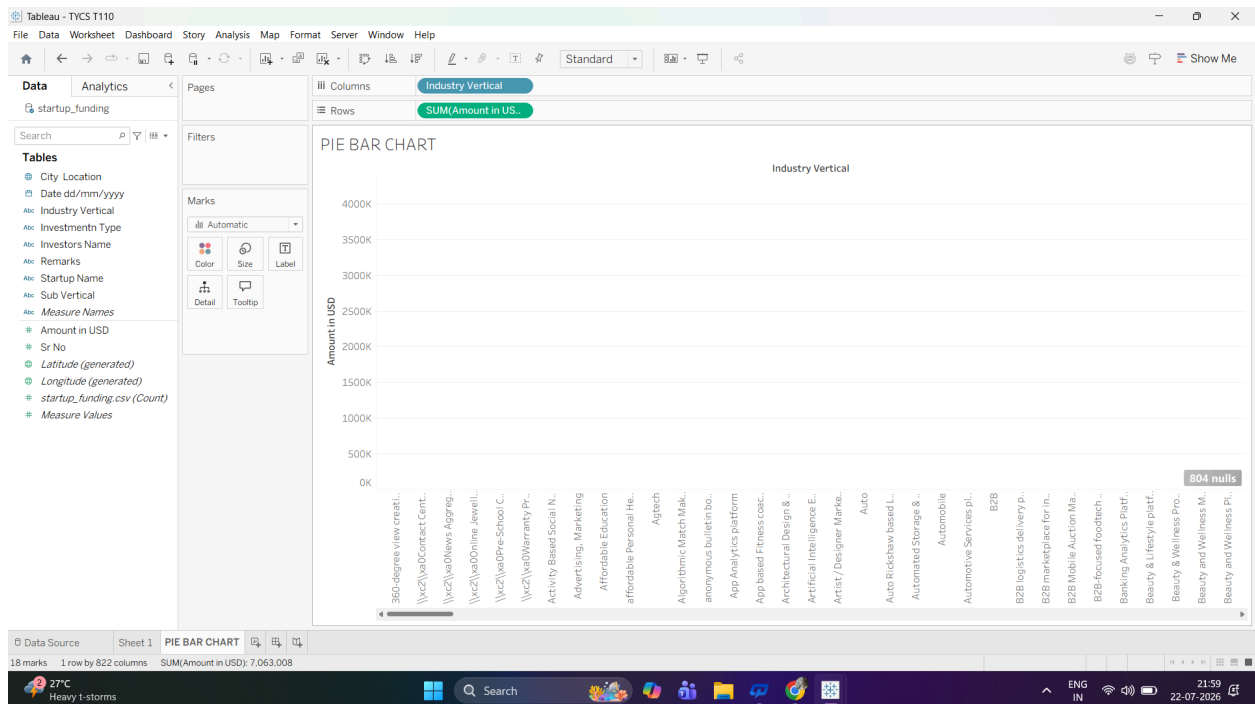
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Step B: Building Treemaps and Pie Charts for Part-to-Whole Analysis

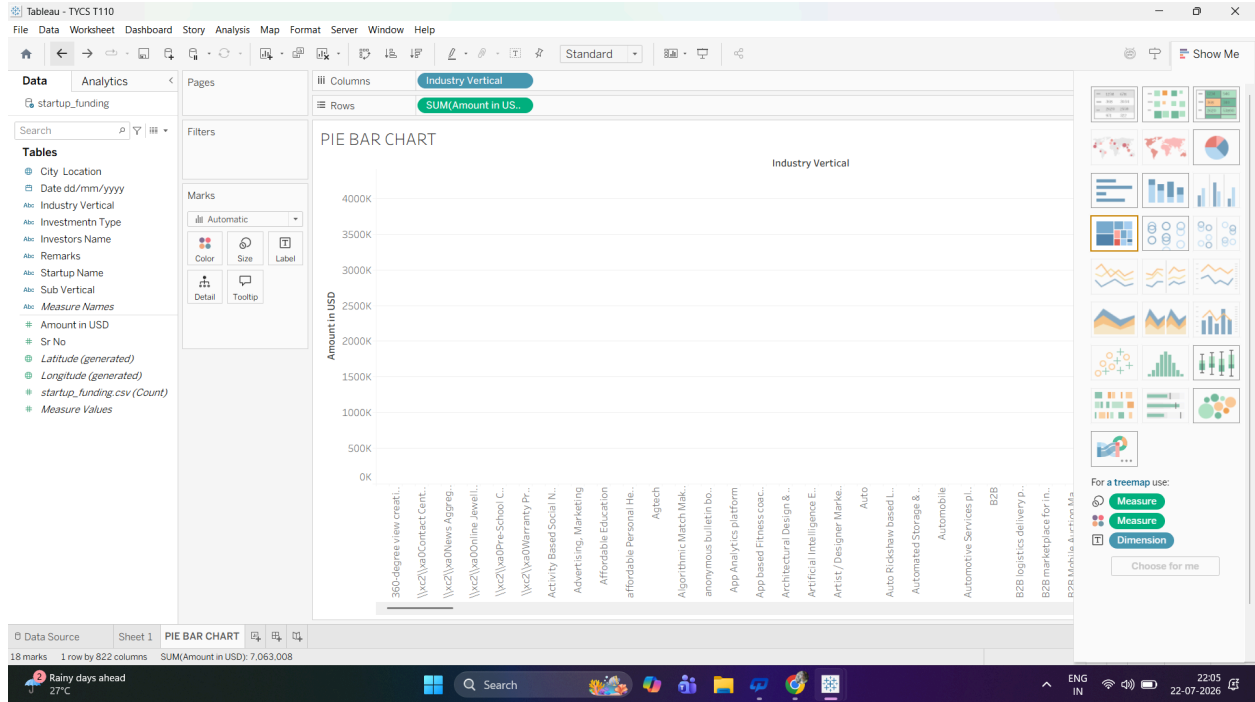
1. Building a Treemap

1. Click the New Worksheet icon at the bottom of the screen to open a clean workspace (Sheet 2).
2. From the Data Pane, drag the Industry Vertical dimension and drop it onto the Columns shelf.

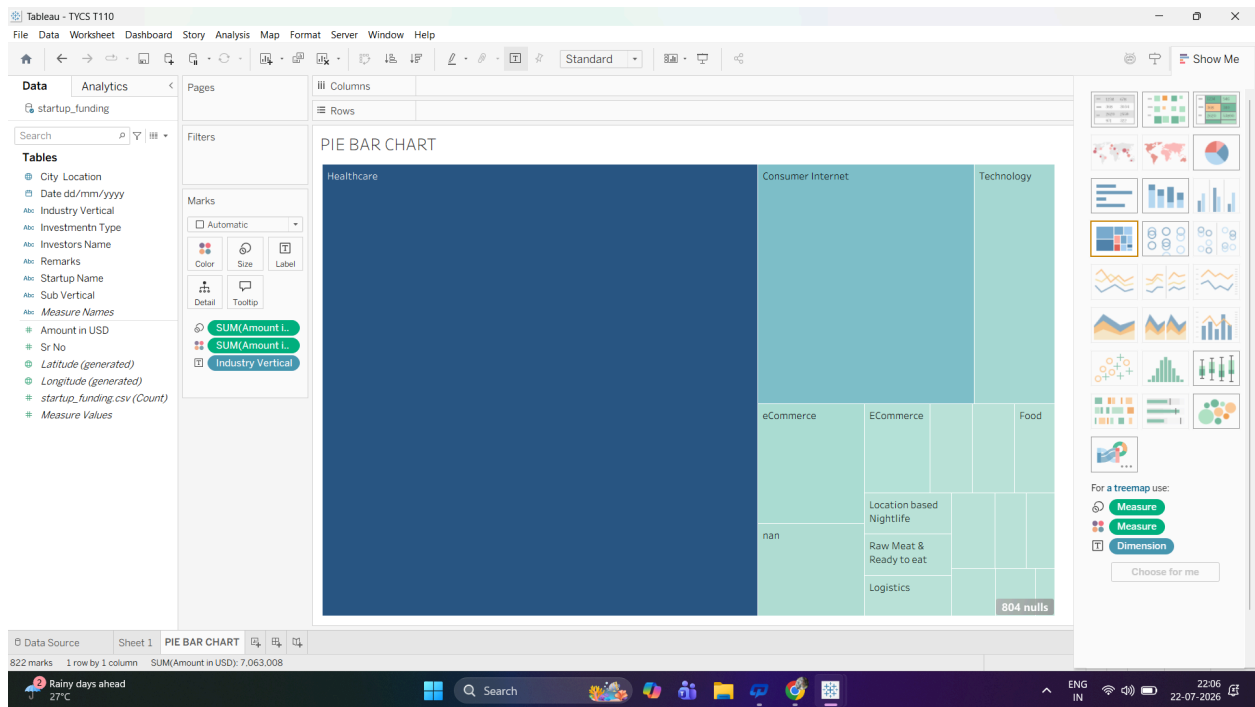


4. Now, look at the very top-right corner of the window and click the Show Me button to expand the chart library panel.

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5. From the options, click on the Treemap icon (it looks like a grid of nested colorful rectangles).

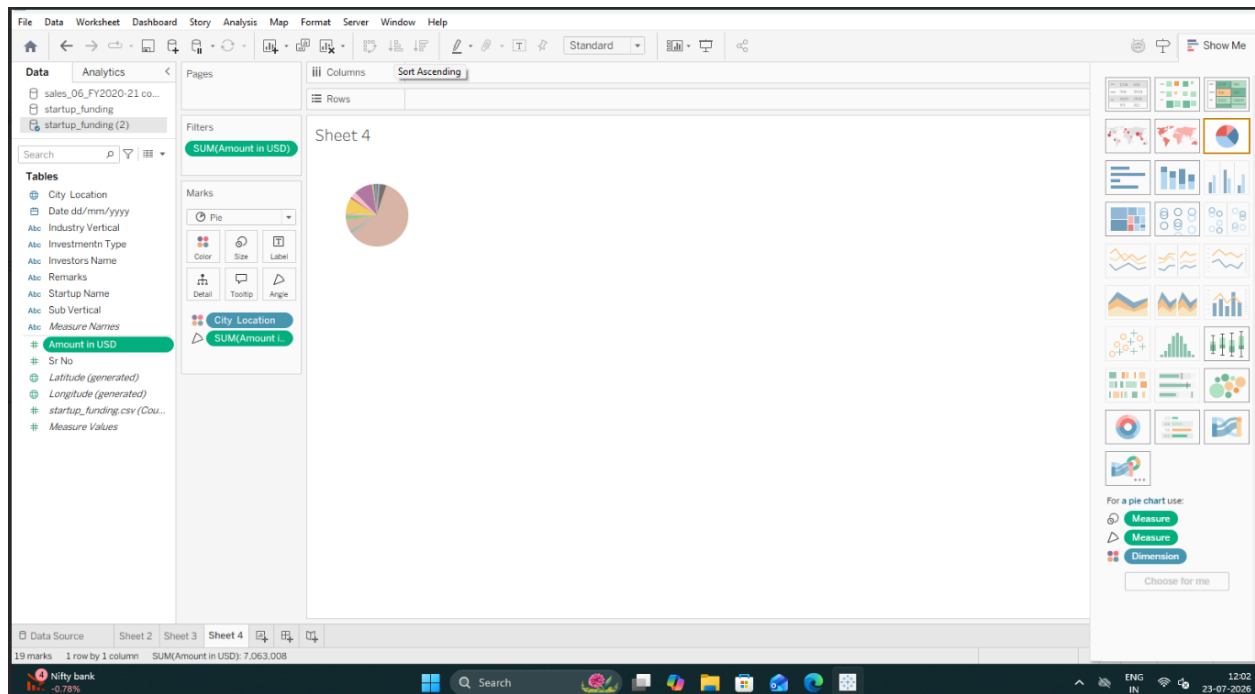


2. Building a Pie Chart

1. Click the New Worksheet icon to open another blank layout (Sheet 3).
2. Go back to the Show Me panel in the top-right corner and select the Pie Chart icon.
3. Notice that your Marks card changes to show an angle option. From the Data Pane, drag InvestmentnType and drop it onto the Color tile in the Marks card.

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4. Drag Amount in USD from the Data Pane and drop it directly onto the Angle tile in the Marks card.



Step C: Designing Scatter Plots to Identify Relationships

1. Cleaning the Data for Scatter Plots

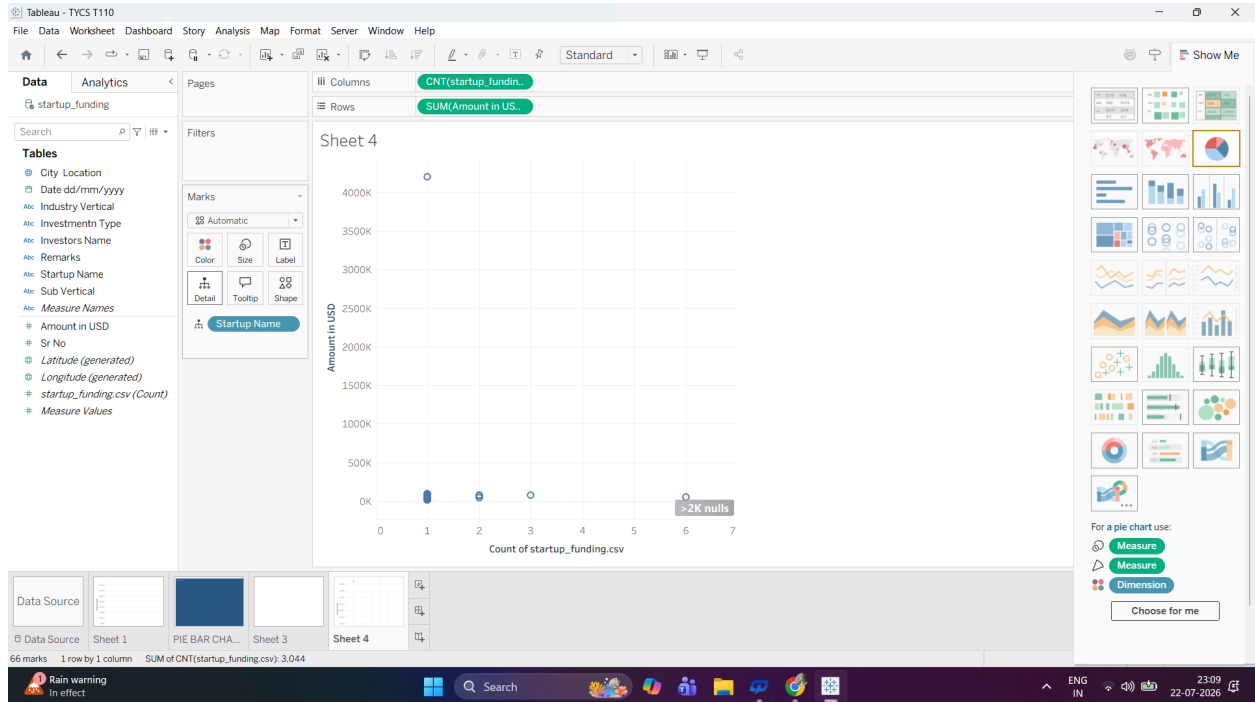
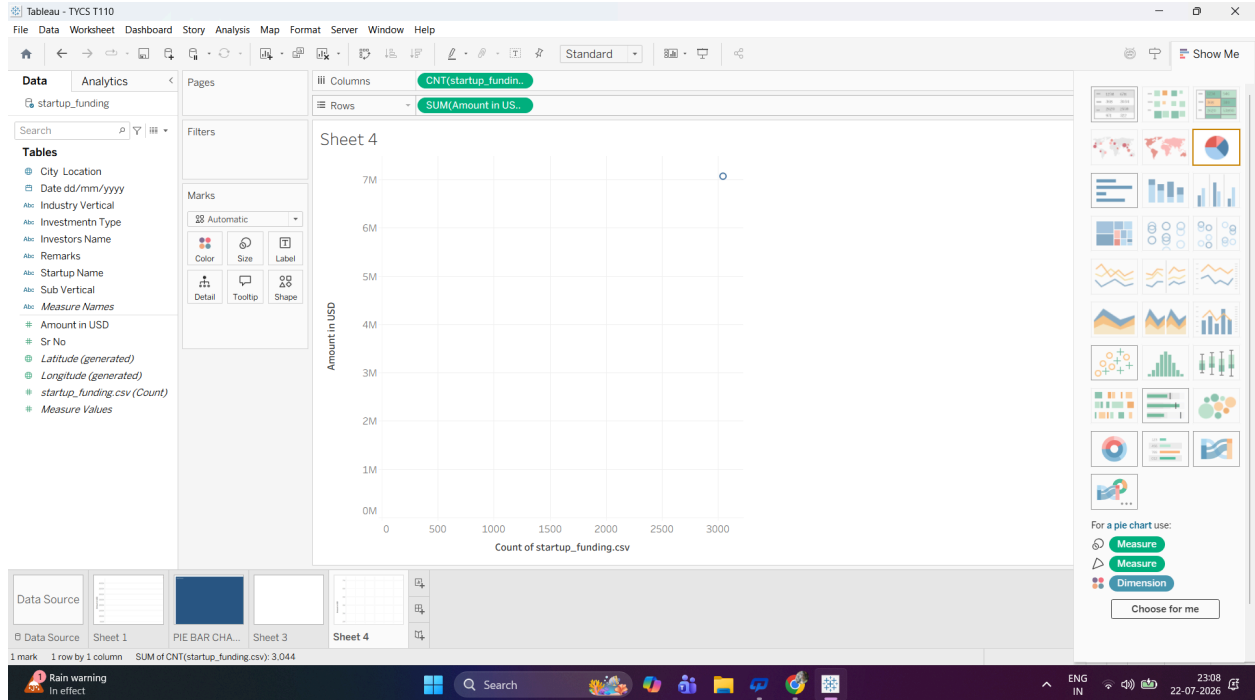
A scatter plot maps relationships between two different numerical measures. However, in this dataset, we only have one explicit primary numeric field: Amount in USD.

To create a meaningful scatter plot, we will instruct Tableau to generate a second numerical measure dynamically by counting the number of times a startup name appears (representing the total number of funding rounds a startup went through).

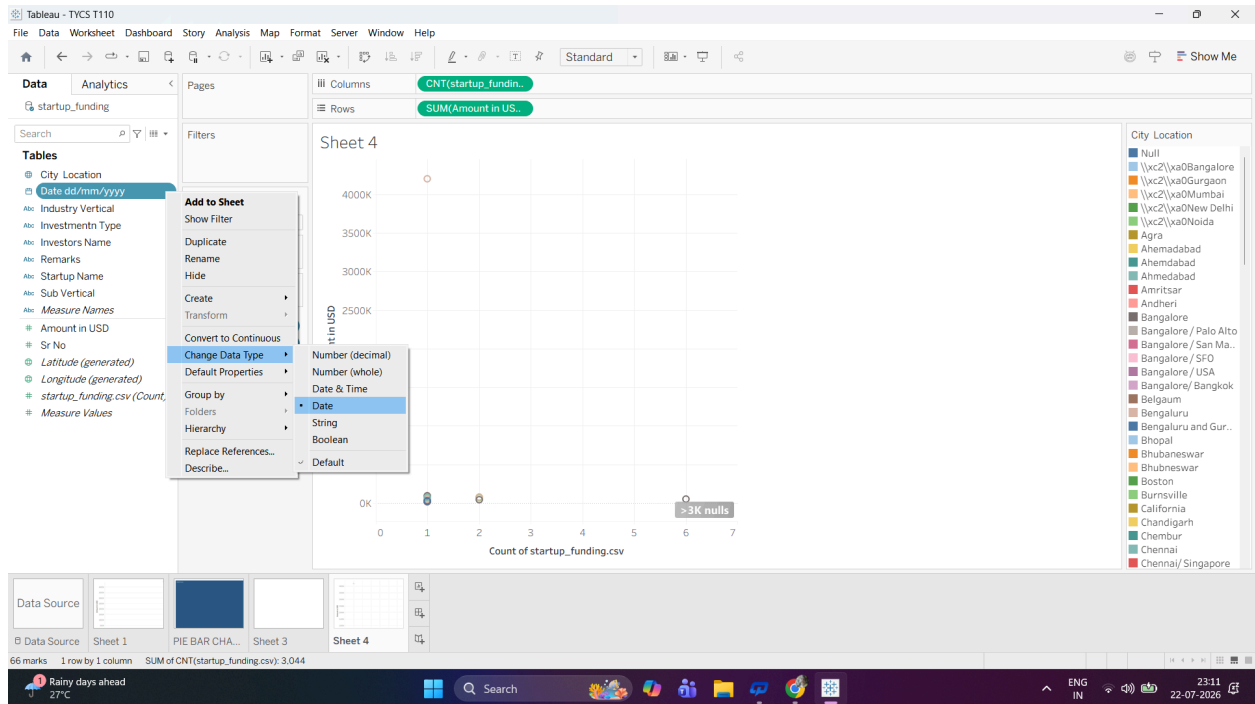
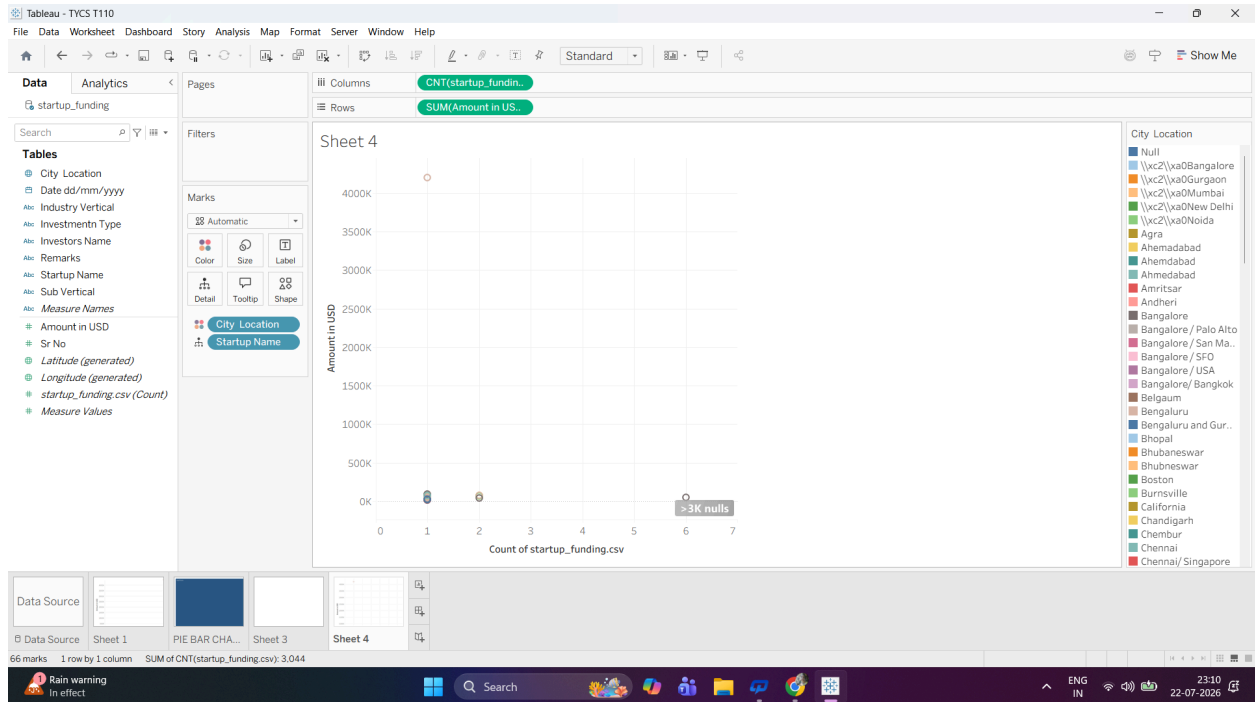
2. Building the Scatter Plot

1. Click the New Worksheet icon at the bottom of your window to open a fresh canvas (Sheet

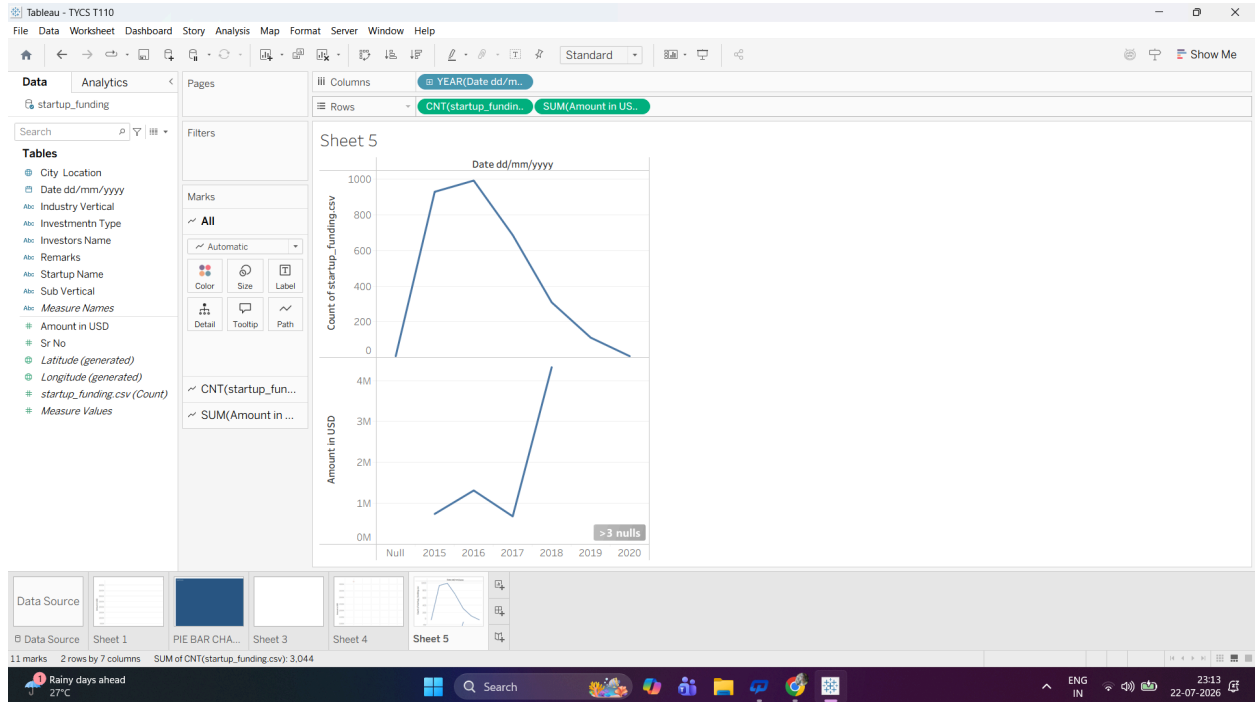
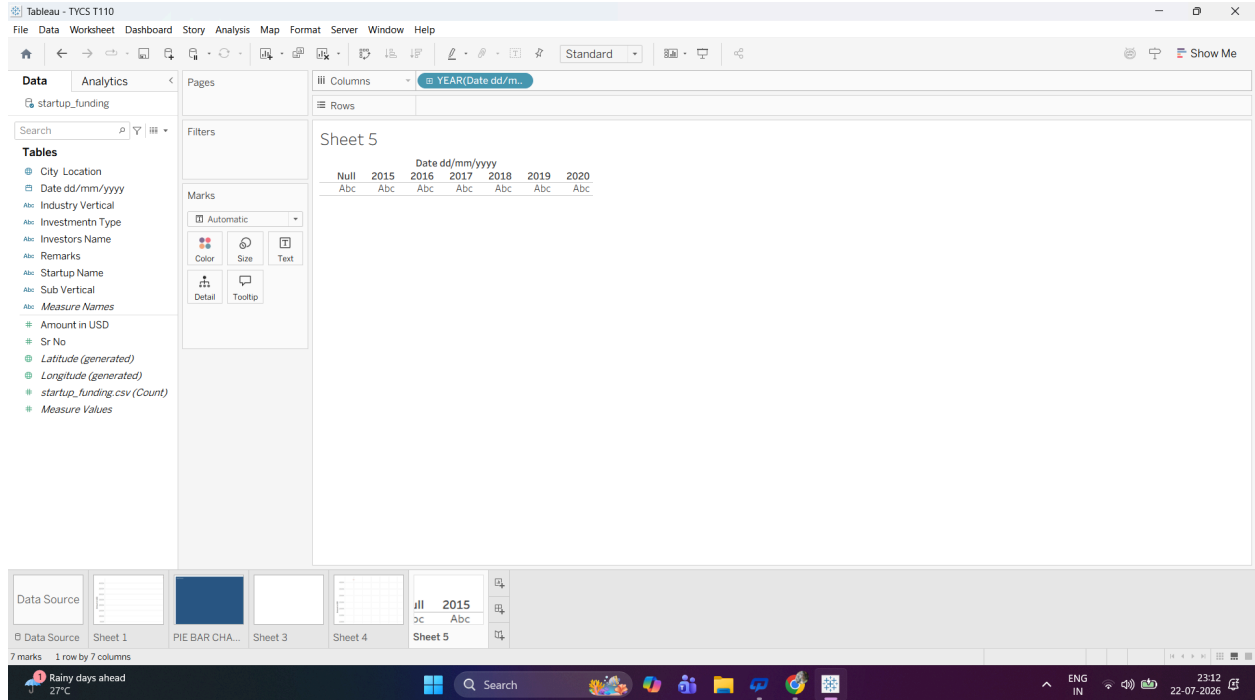
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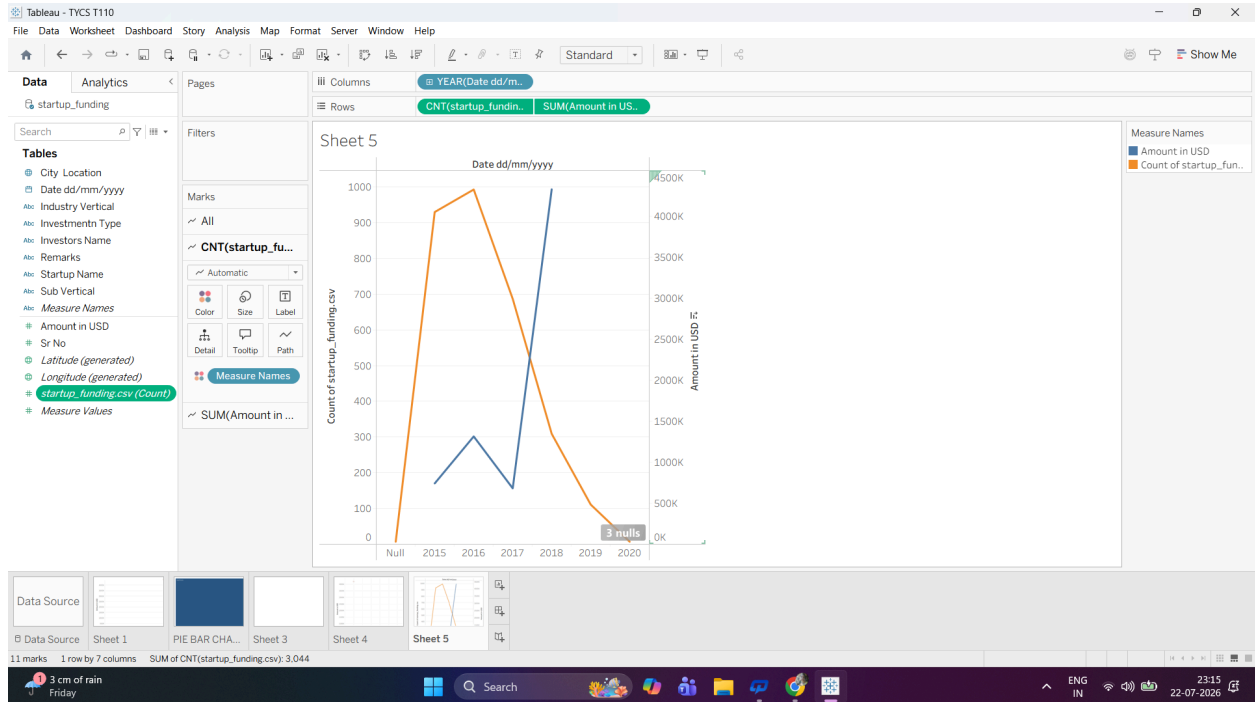
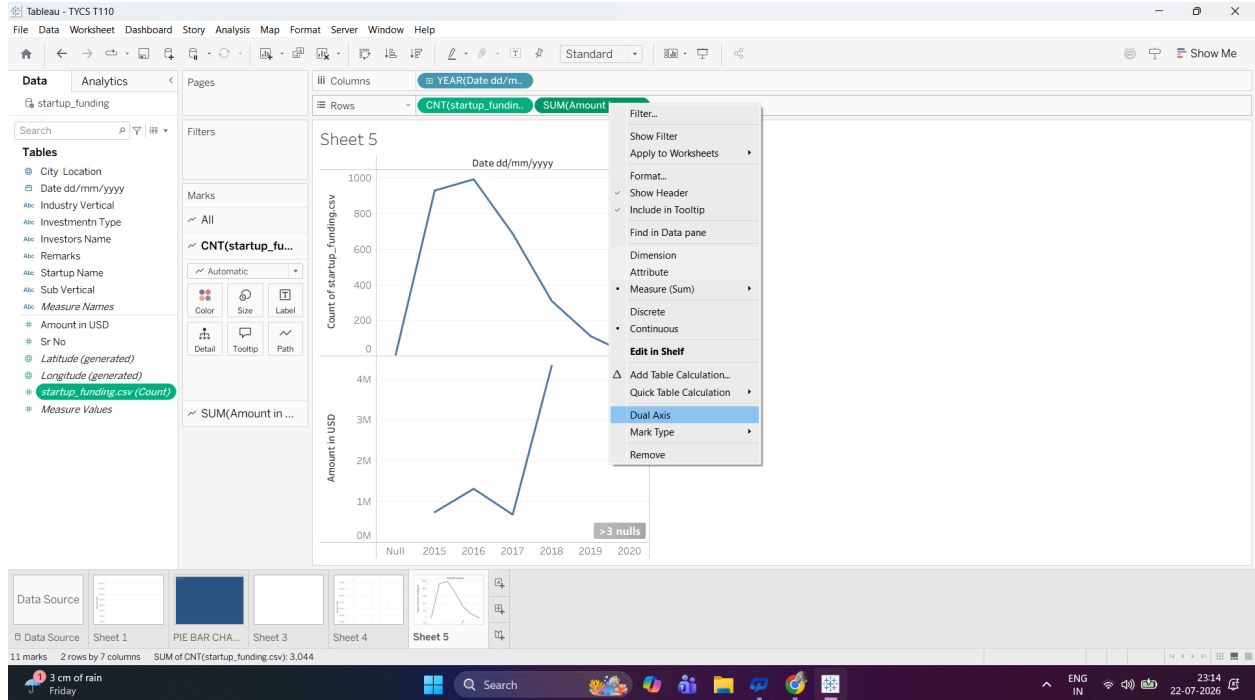
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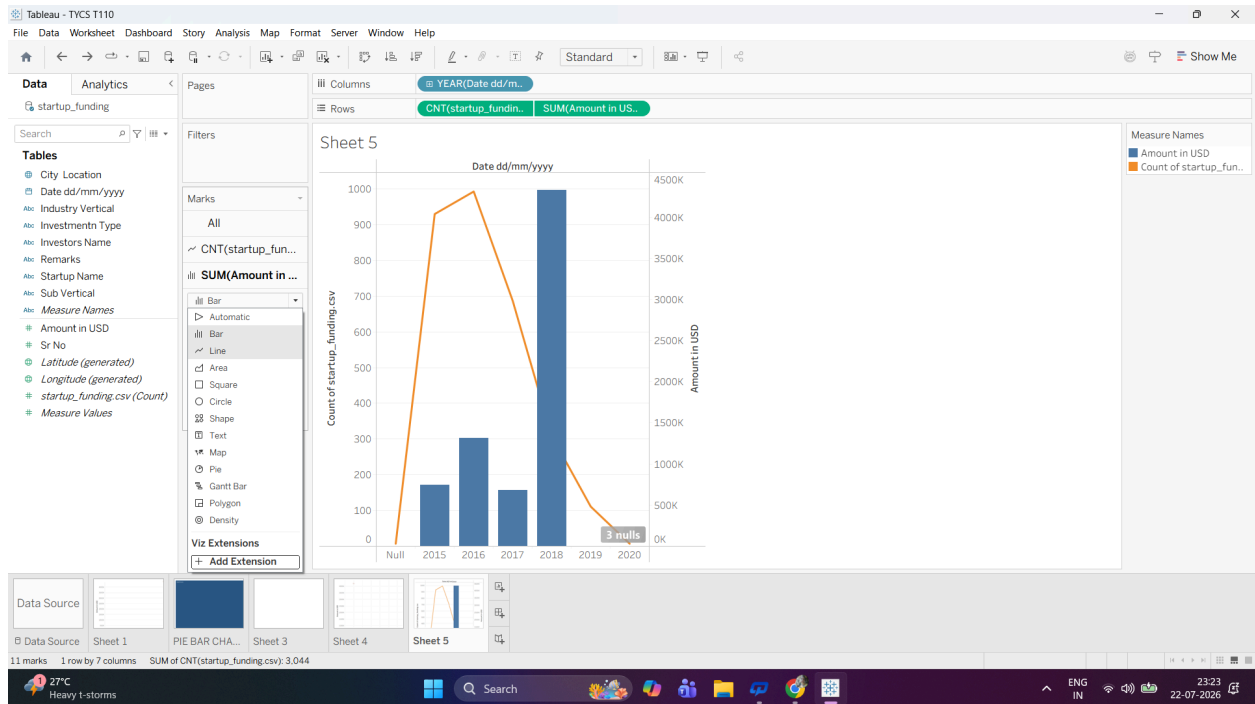
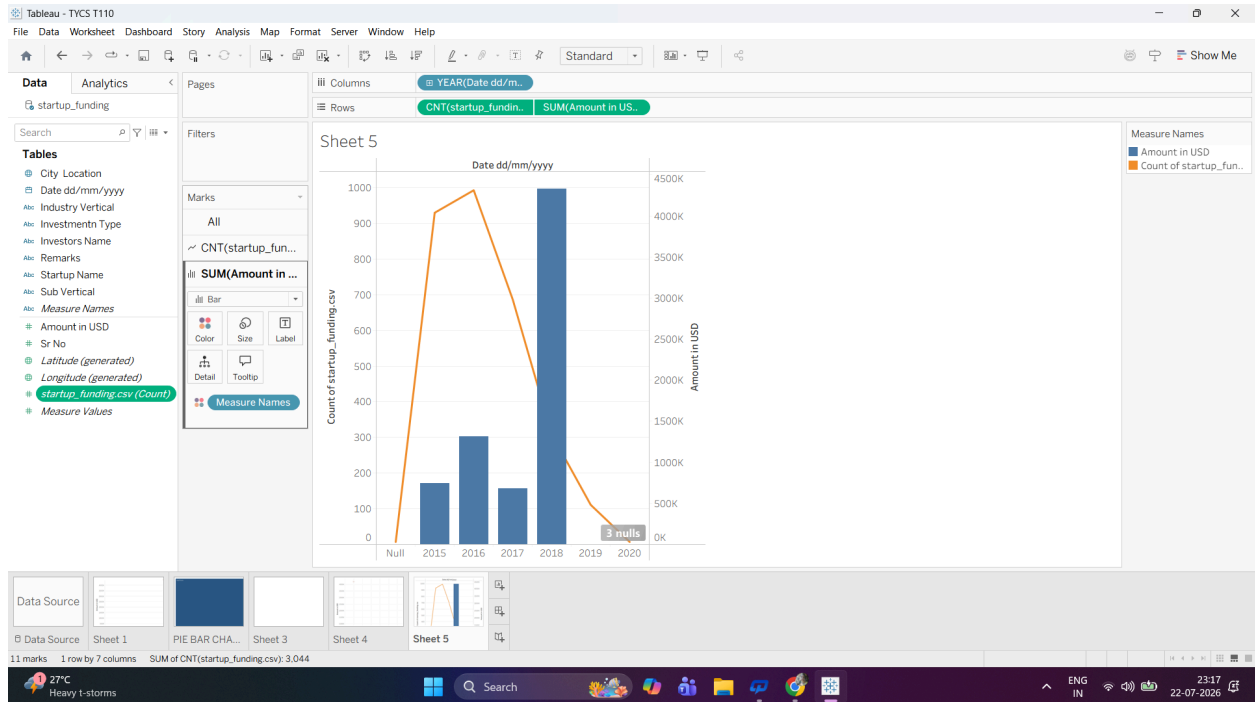
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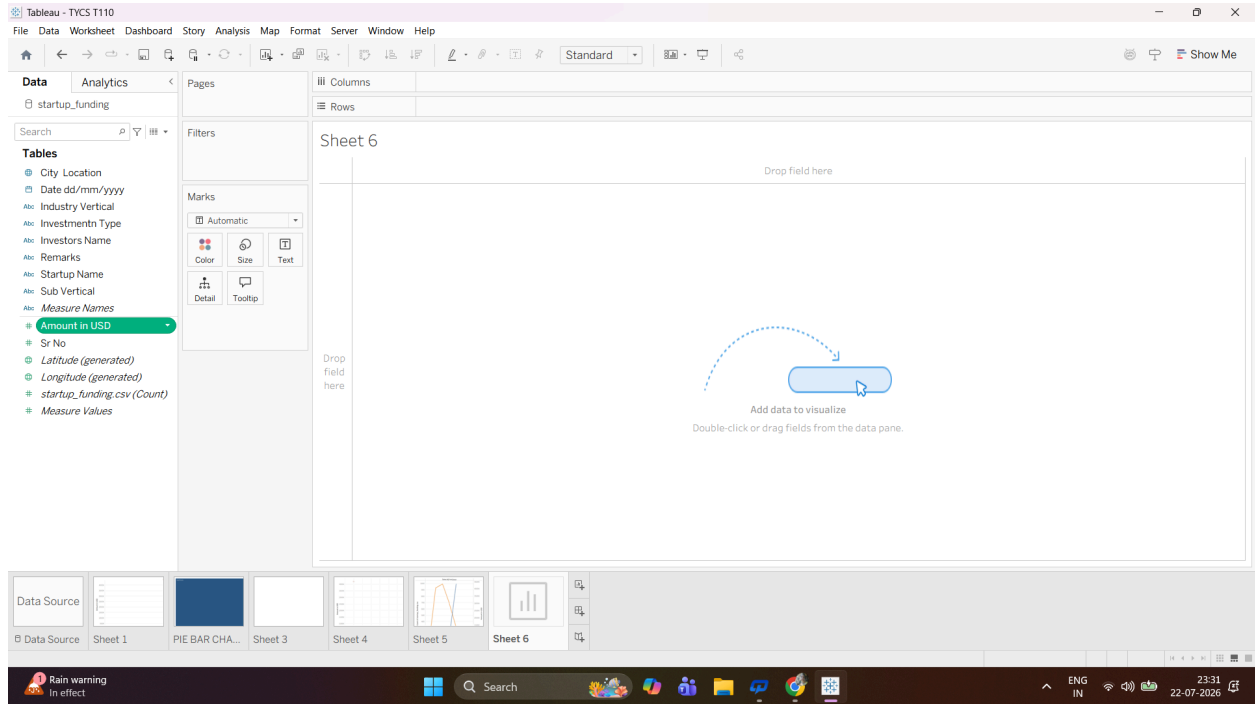
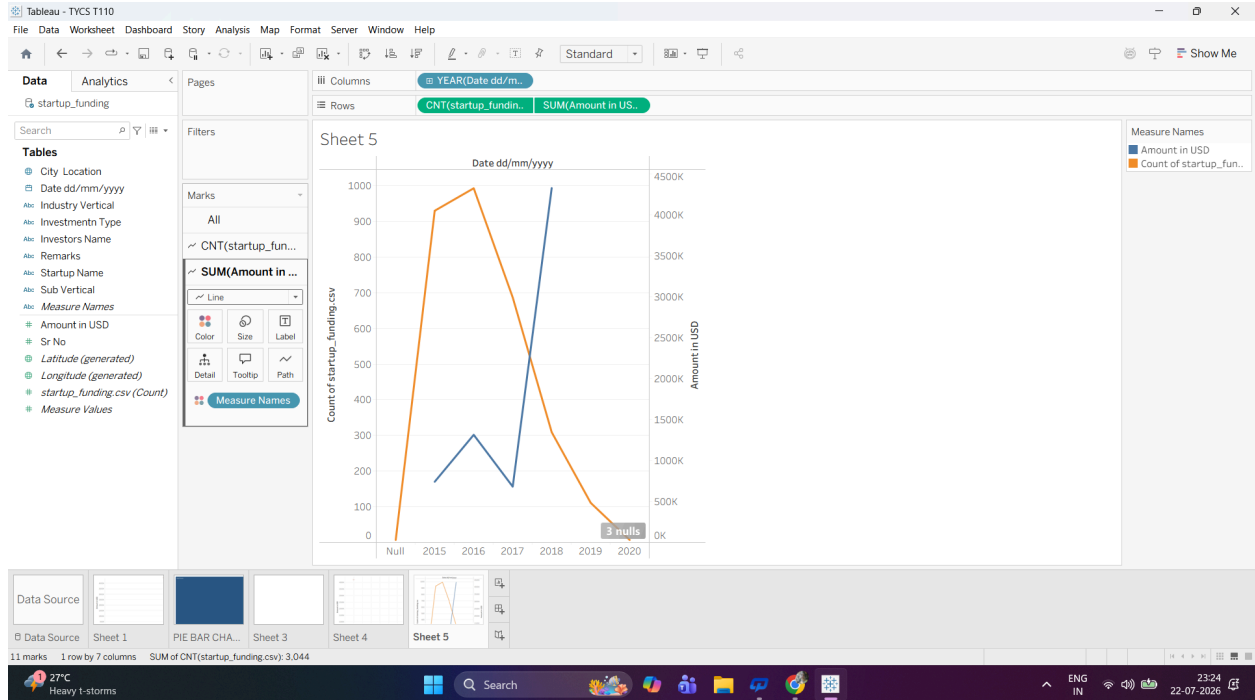
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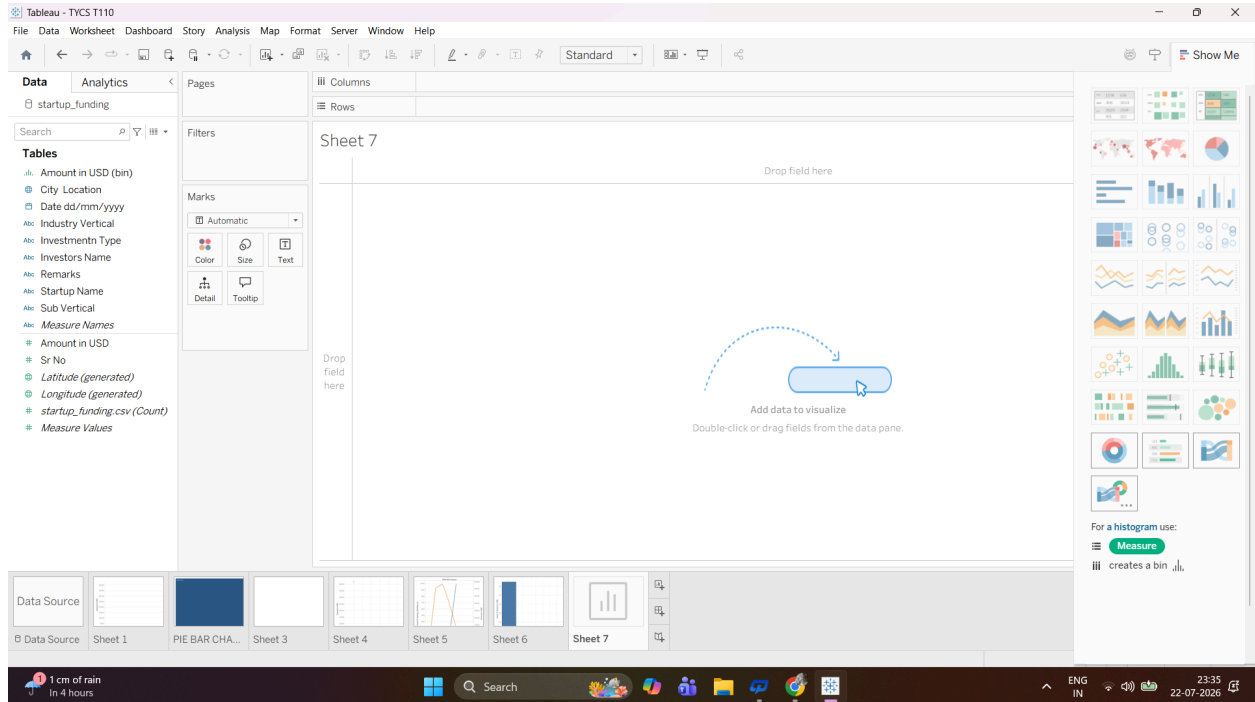
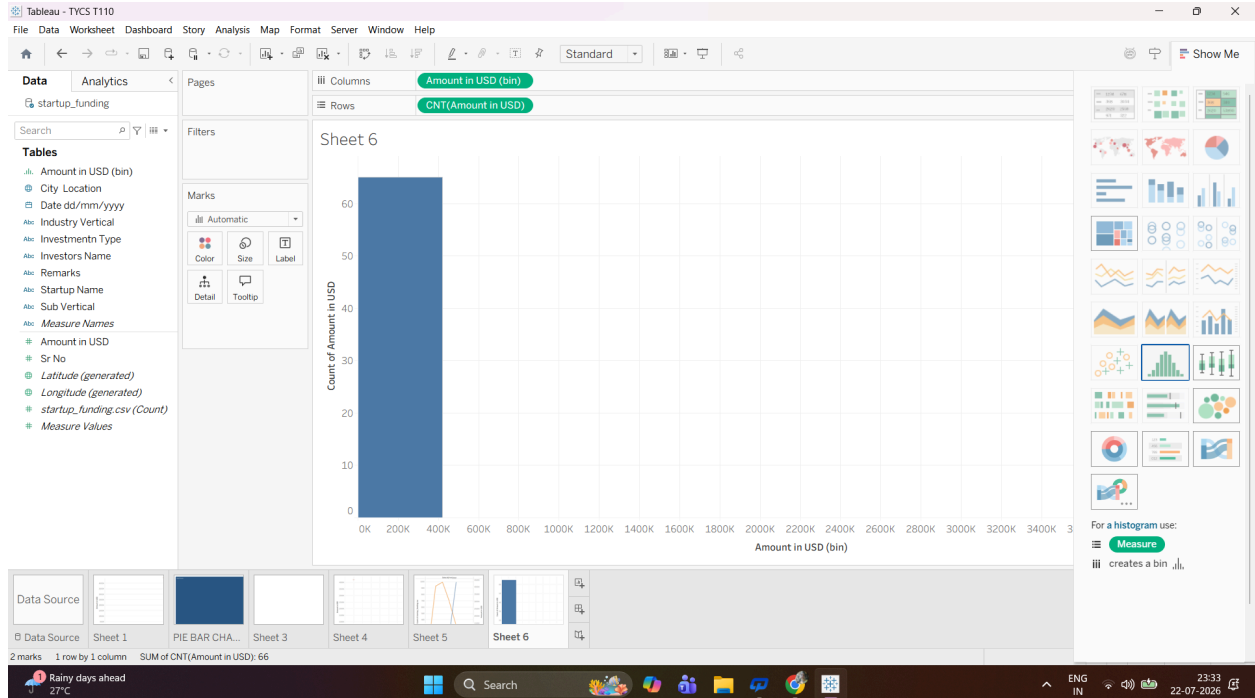
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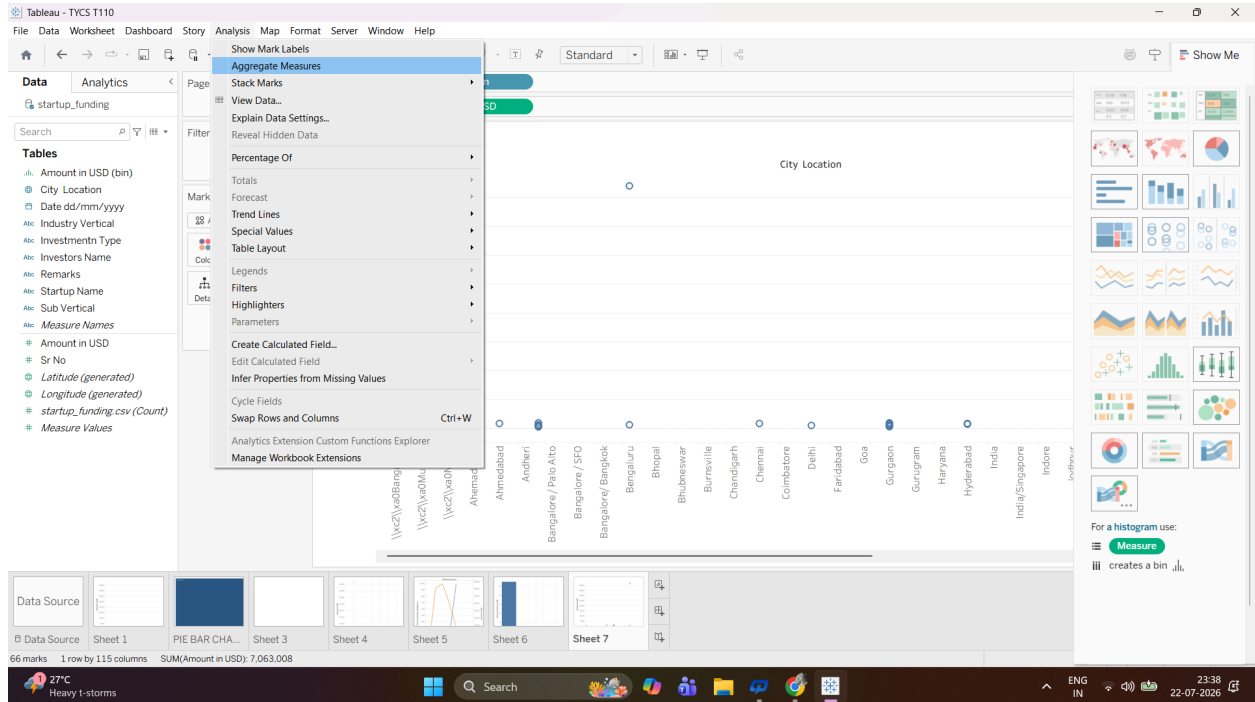
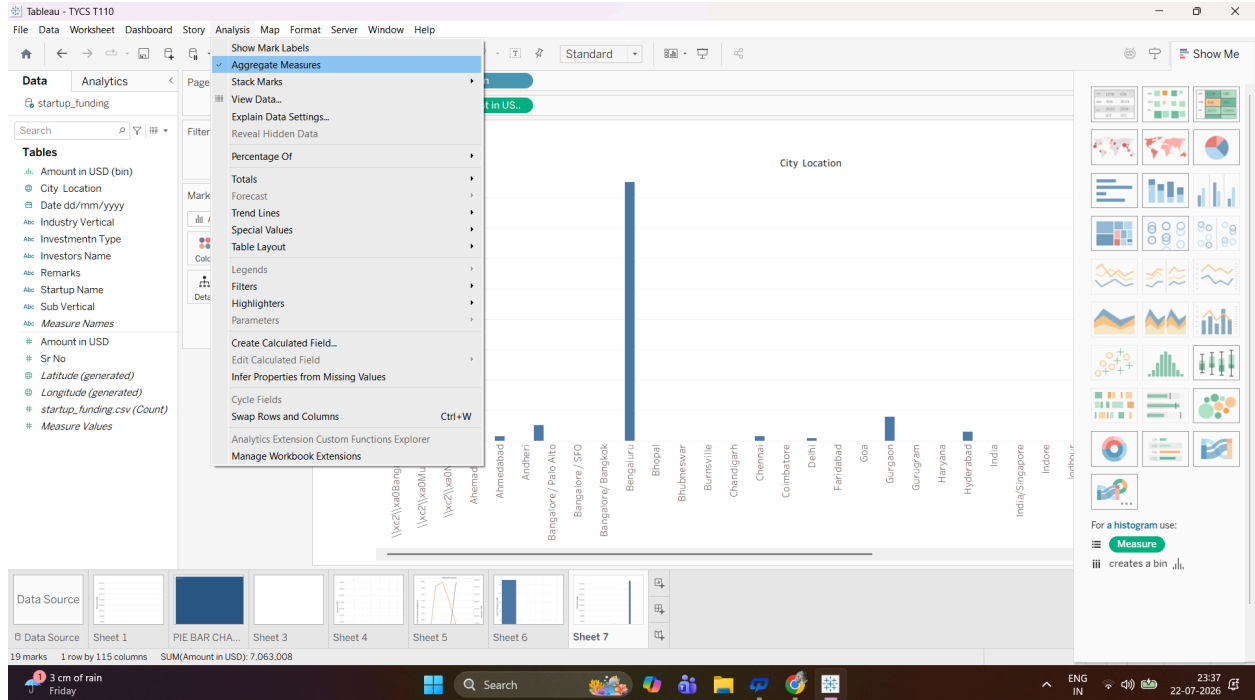


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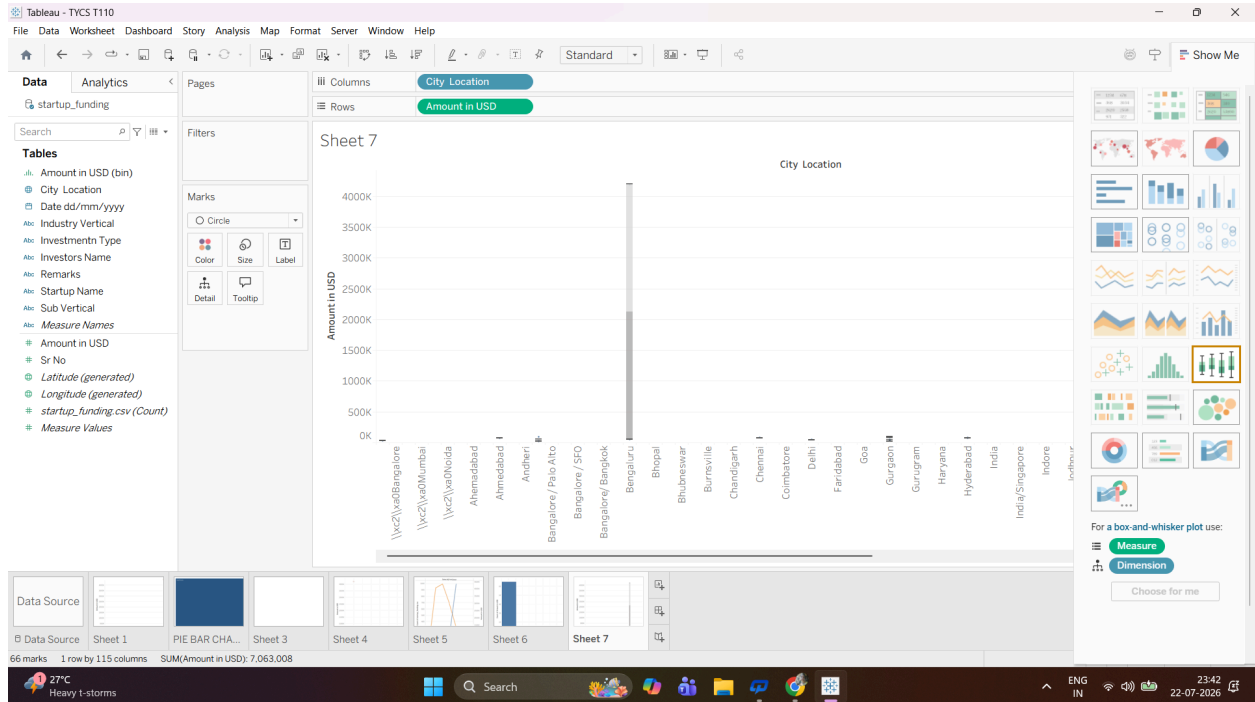
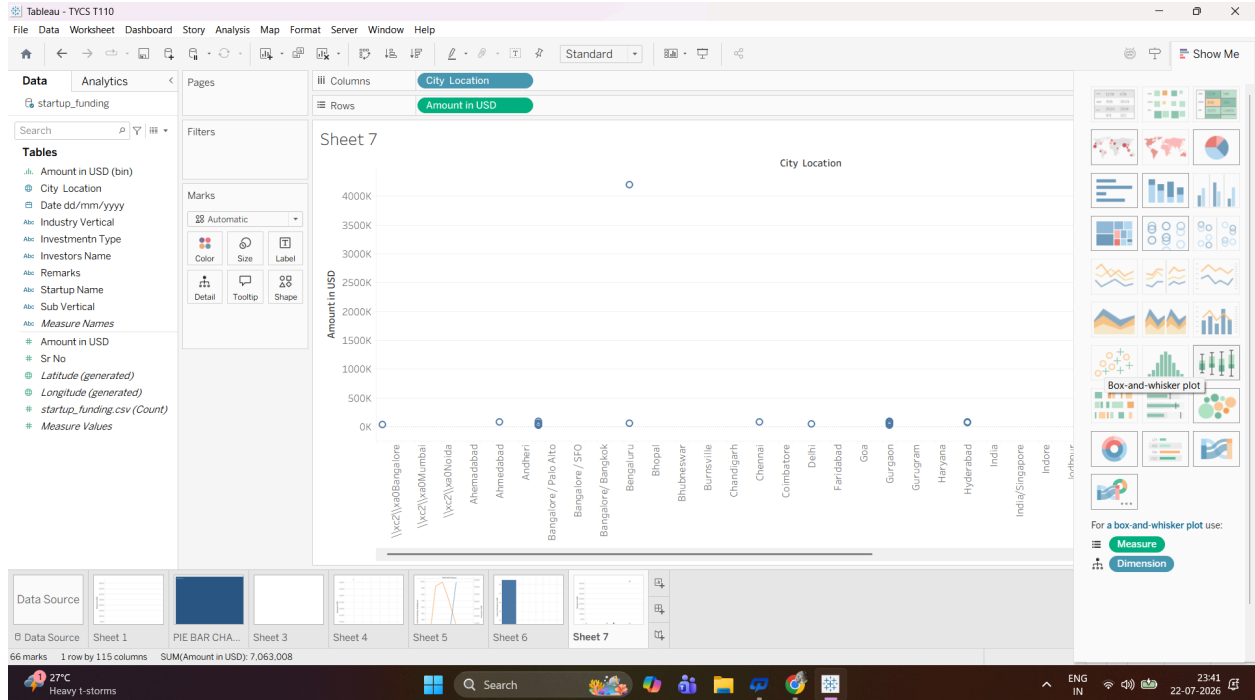


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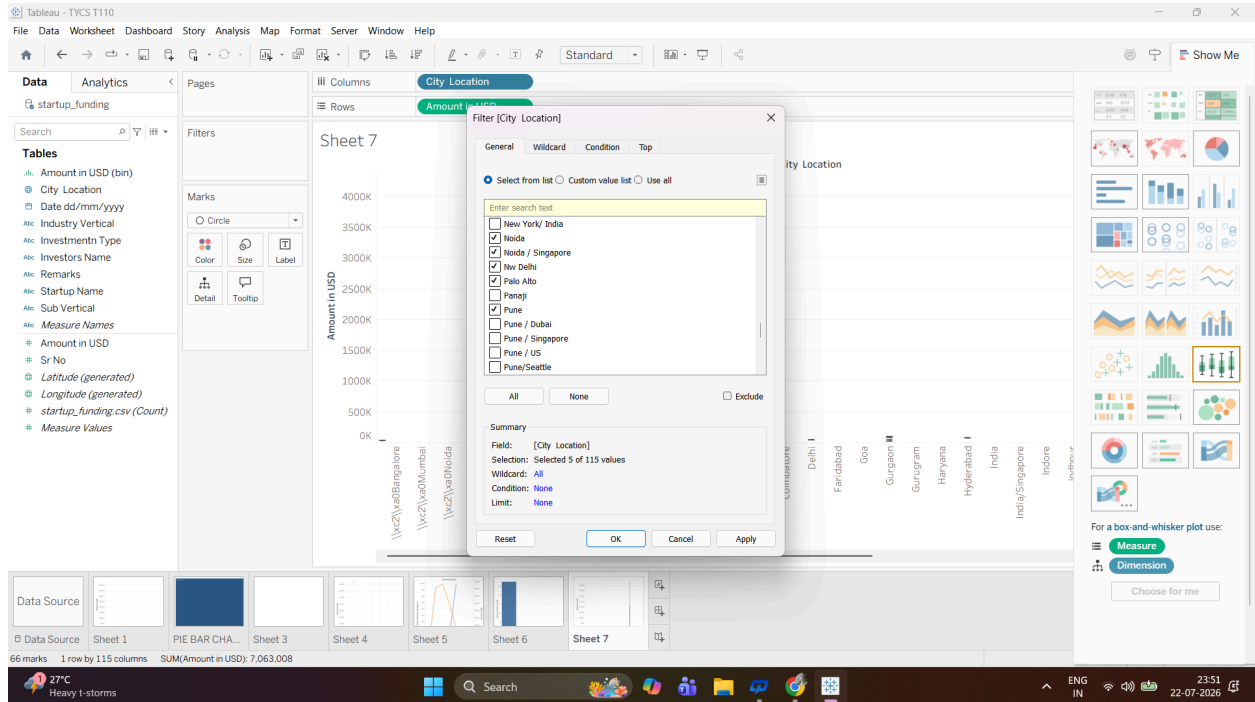
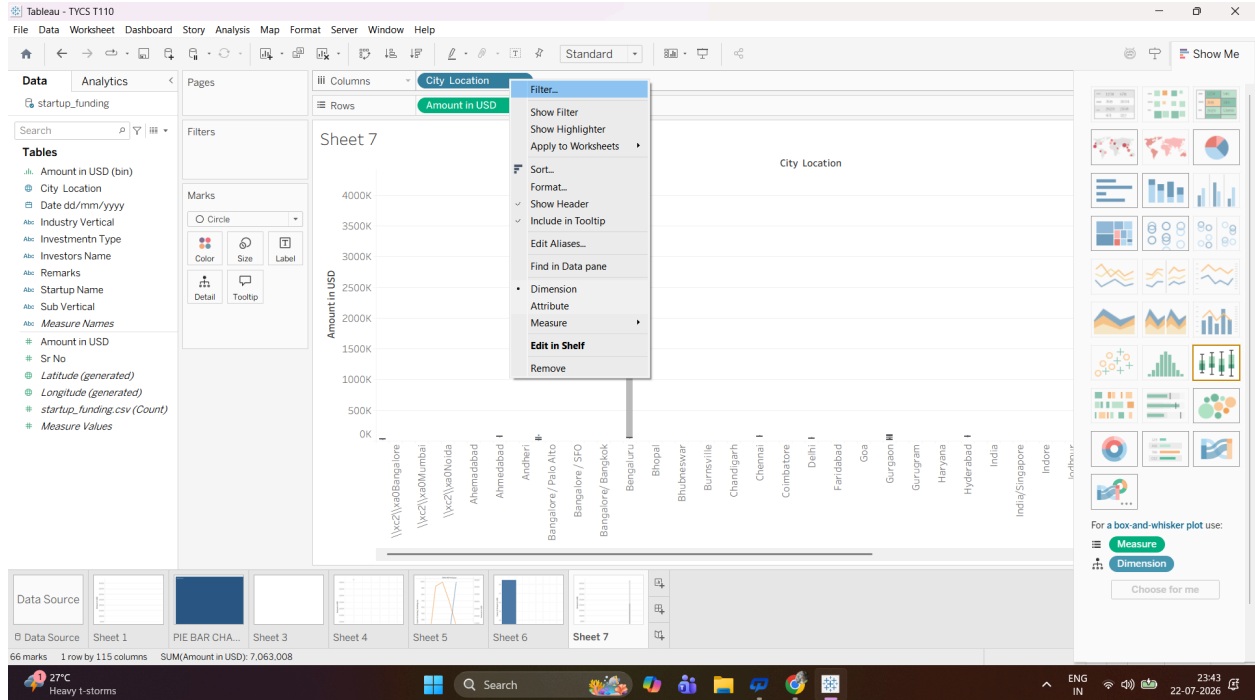
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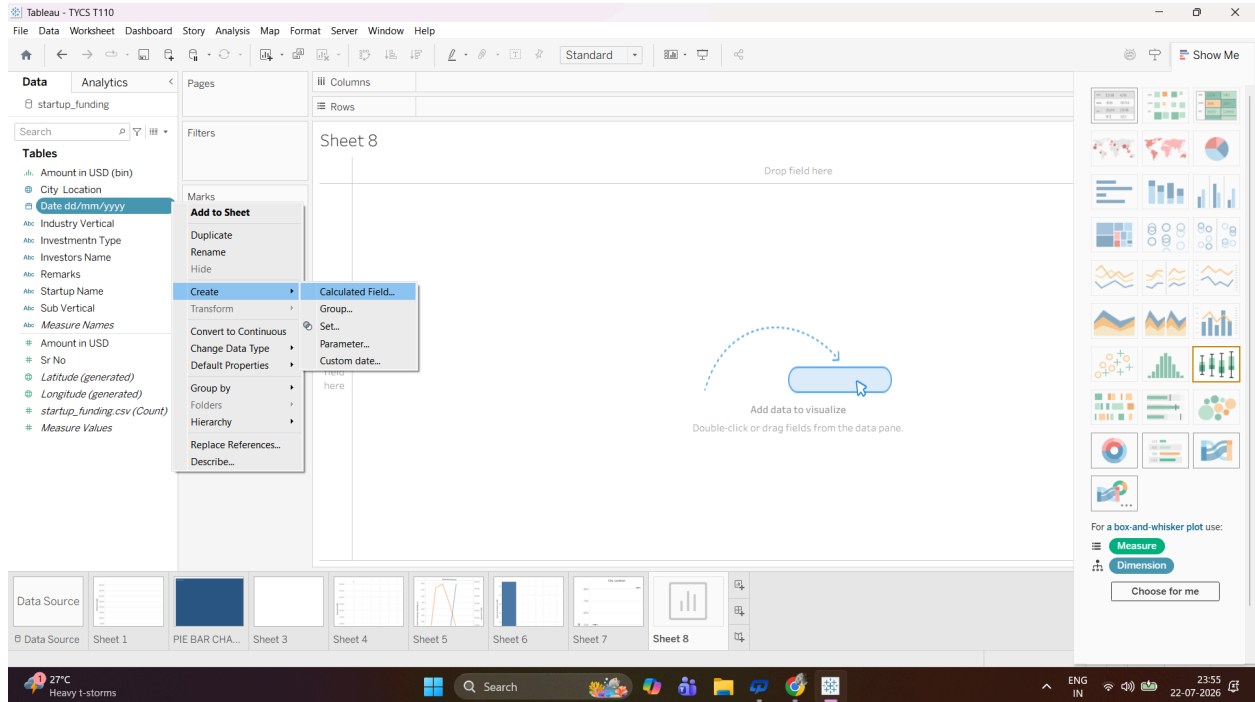
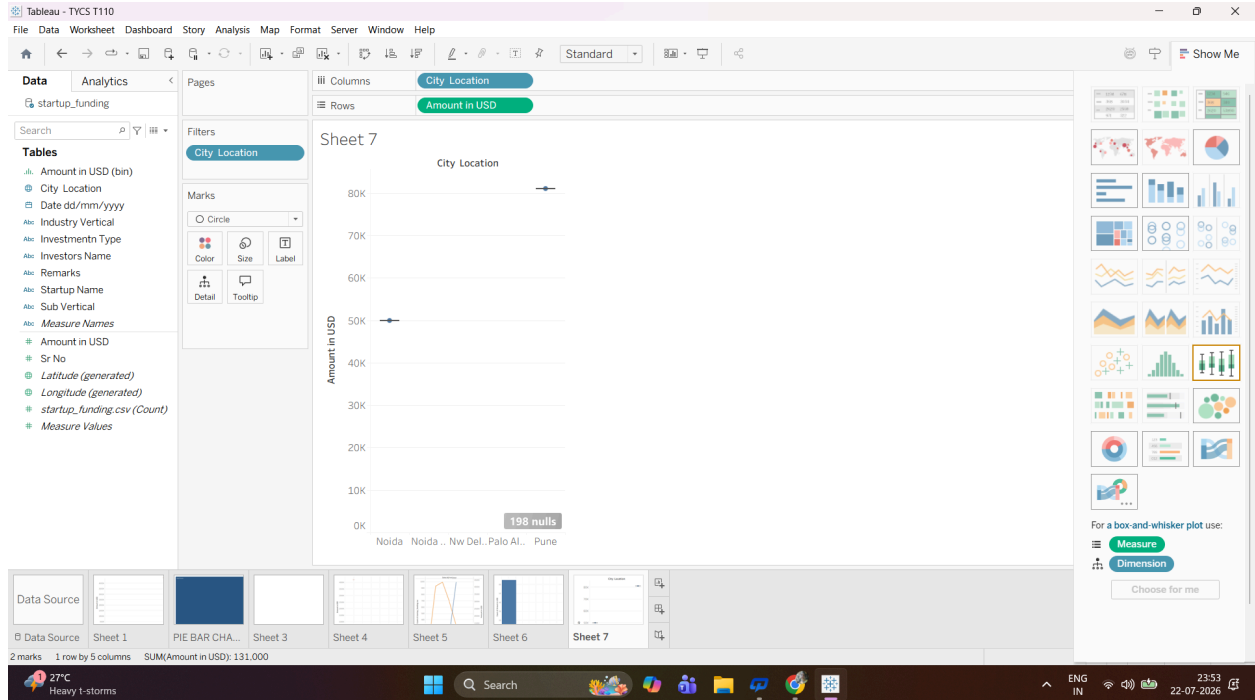
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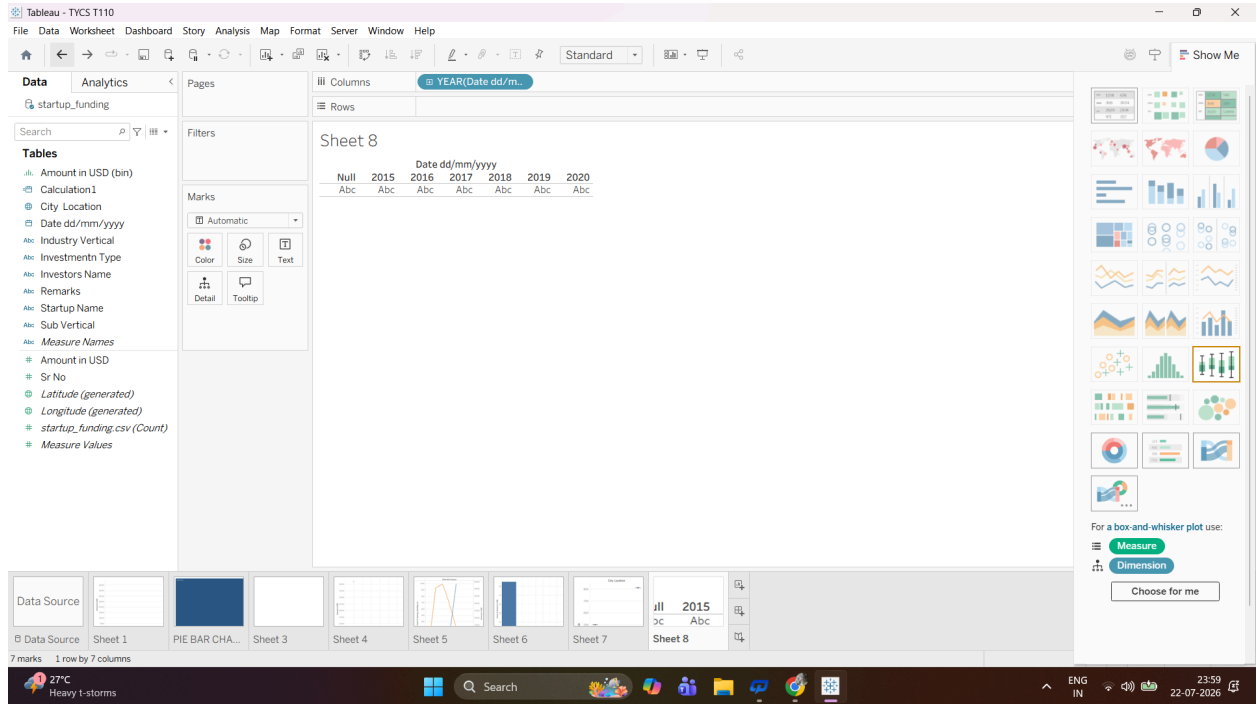
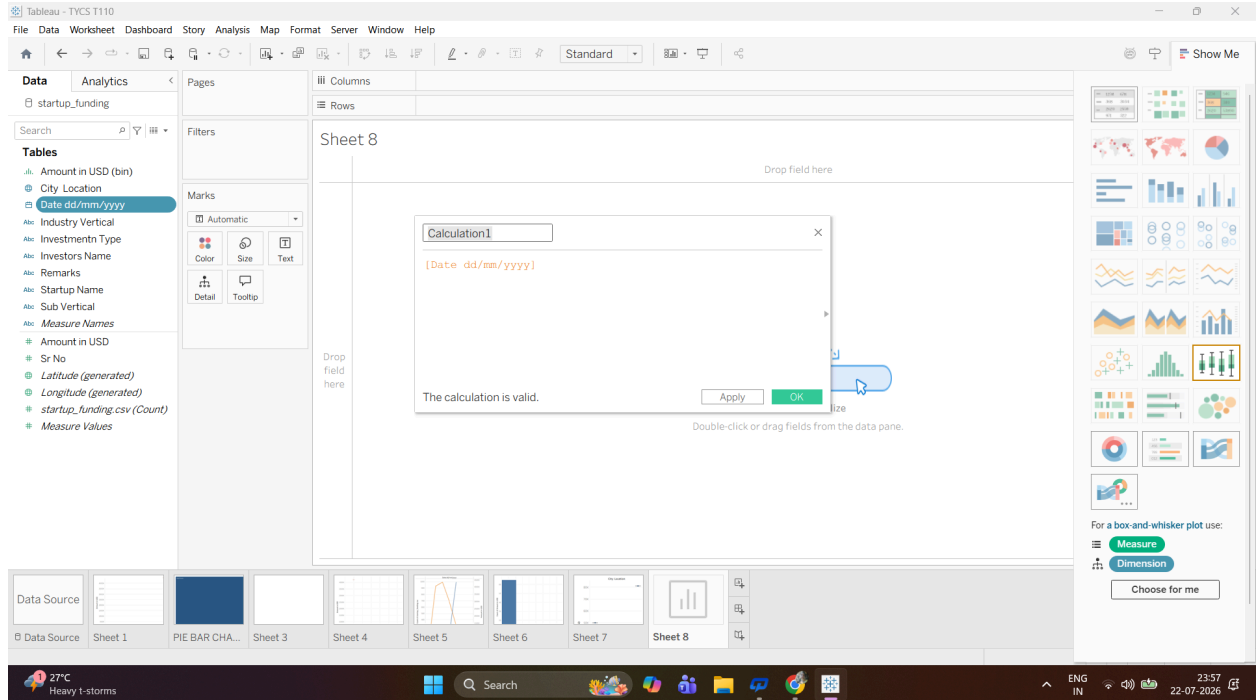
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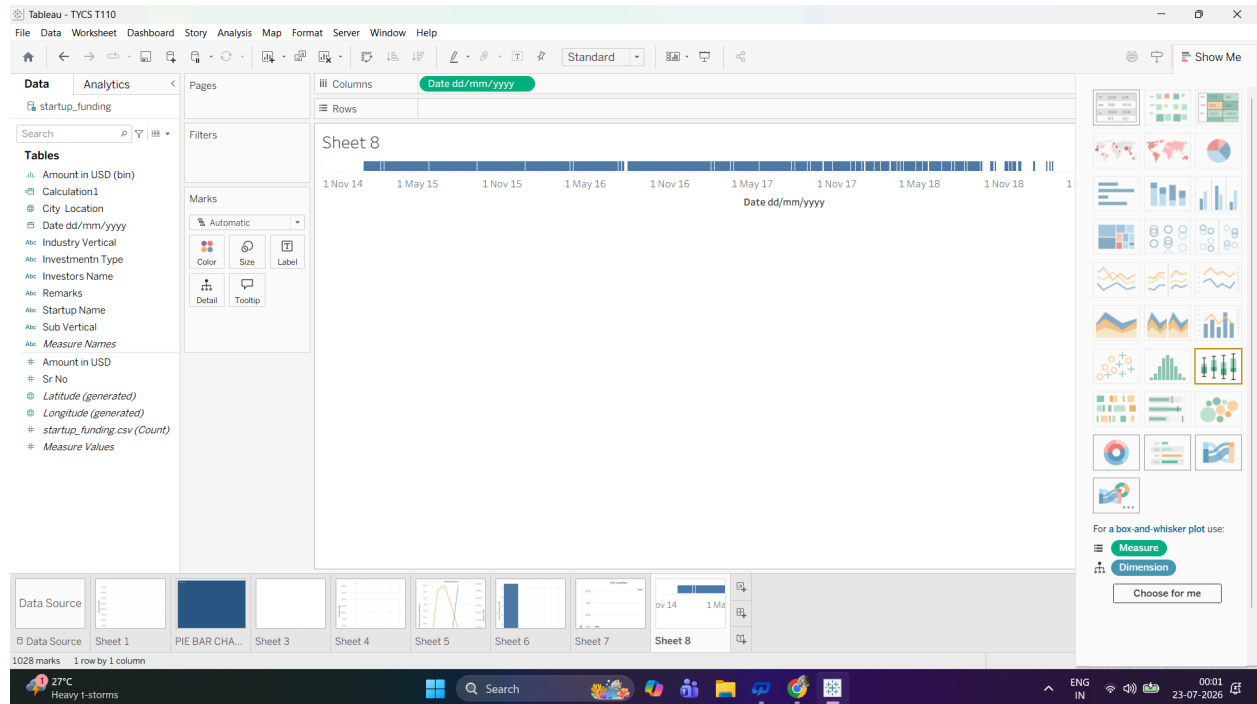
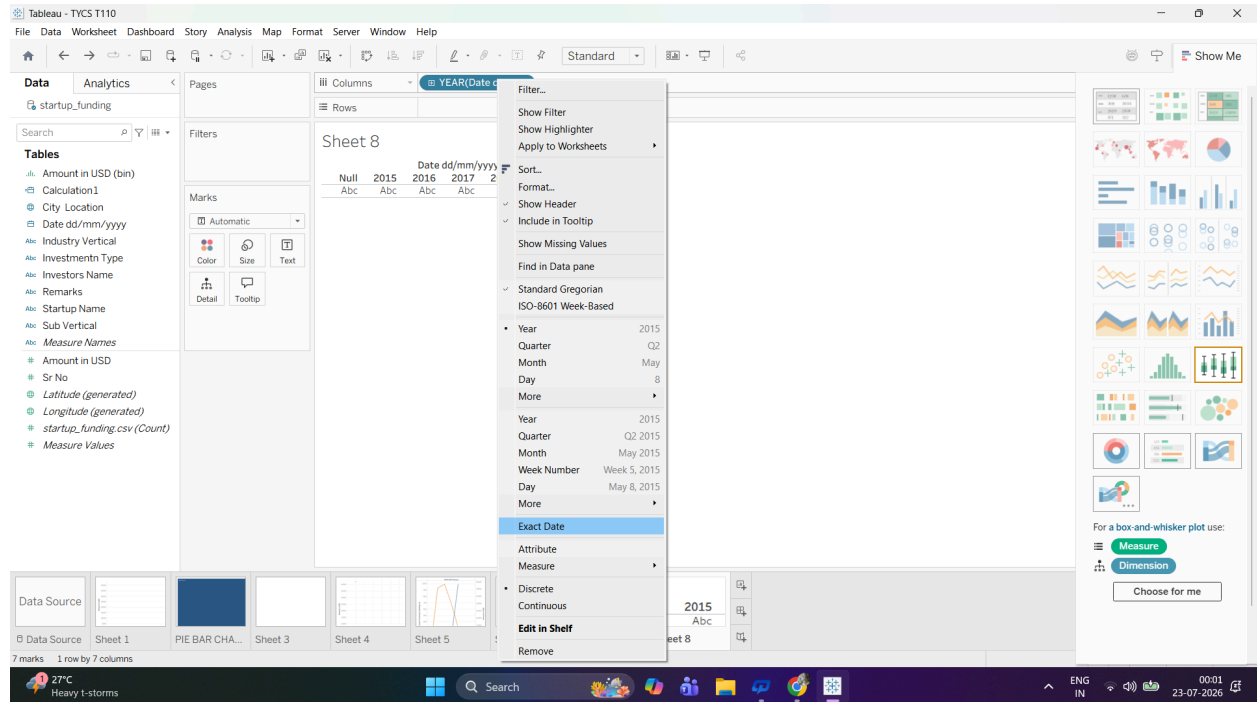
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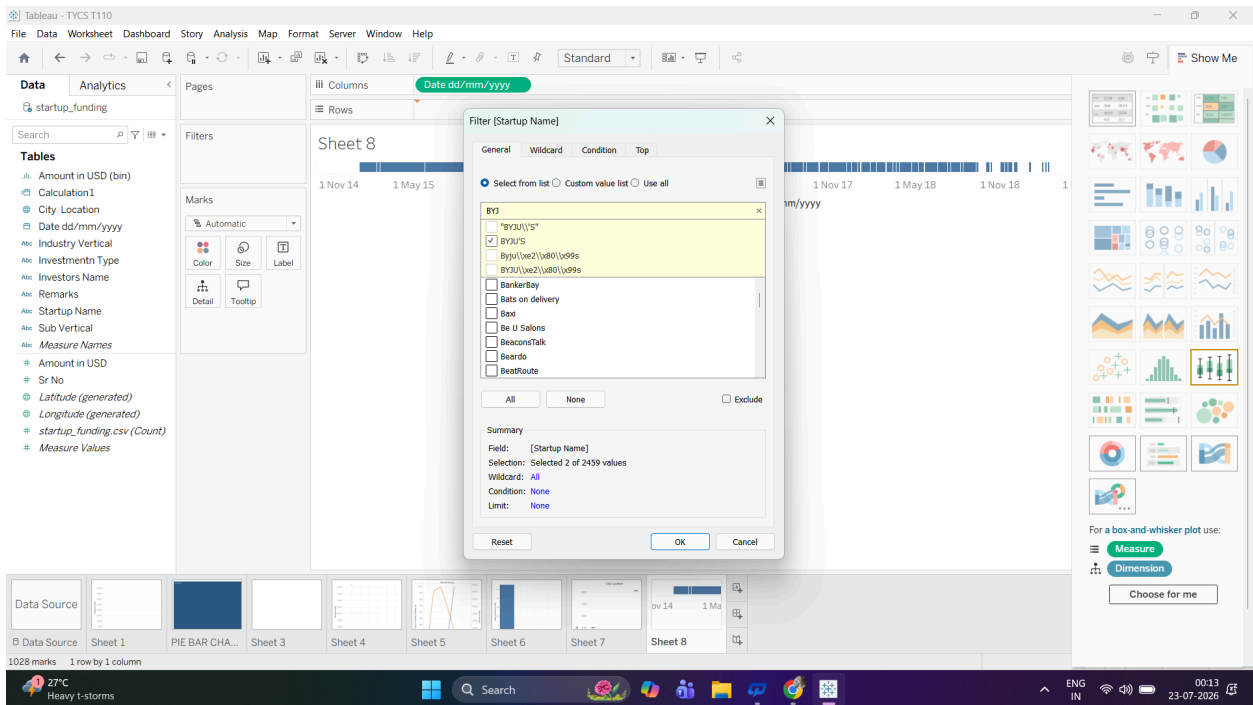
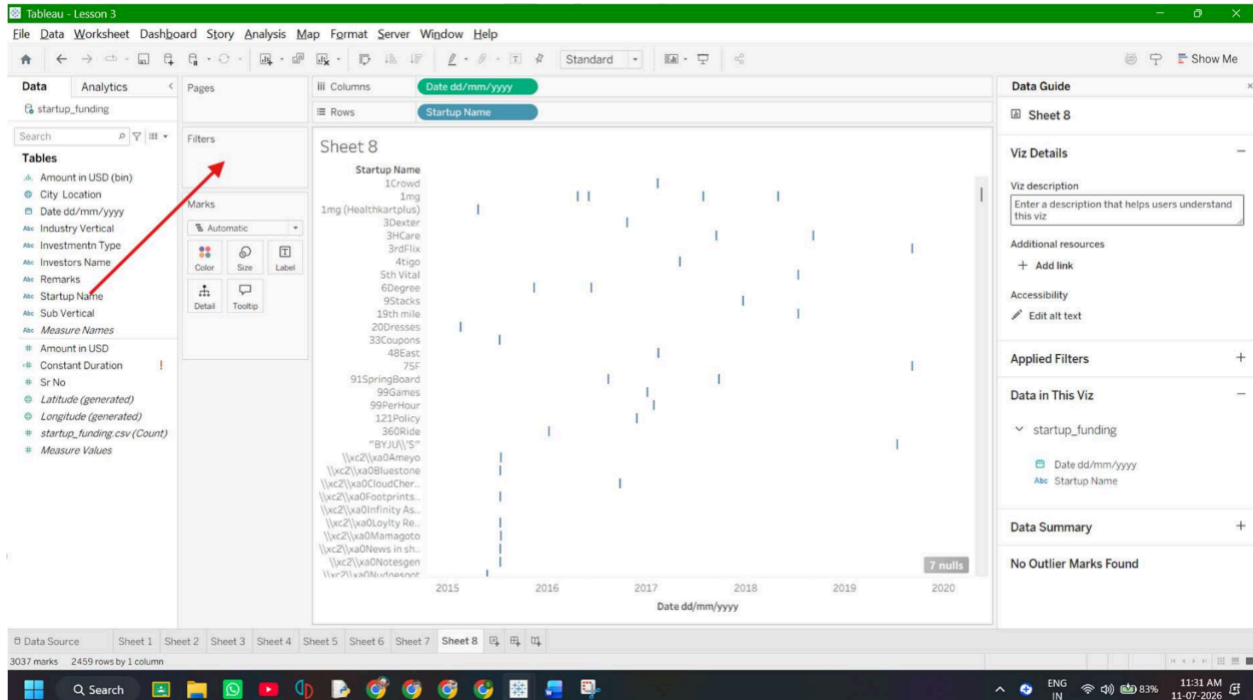
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